An AI-Assisted Cosmological Model Survives Its First Joint Test Across Three Cosmological Probes

Dark energy from growing relations

A Mathematical Materiality proposal explained step by step, with examples, observations and mathematics that can be read separately.

A cosmological model built on an unusual idea has passed its first joint test against real data. Here dark energy is the effect of a network of relations that keeps growing, and one growth rule, controlled by three dark-sector parameters, fixes how that component evolves. So constrained a construction could have failed. It held.

Fitted to 1,590 Pantheon+ supernova light curves, 13 DESI DR2 measurements and the Planck 2018 distance priors, its sampled chain reaches a total chi-square of 1420.02, where lower means a closer match; a cosmological constant in the same code reaches 1421.19. The parameter p, which sets the departure from a cosmological constant, now has an upper bound of 0.0724 at 95 per cent credibility.

With three extra dark-sector parameters, this shows viability, not preference, and the microwave background enters only through its compressed geometry. Code, data handling and audit record are deposited for independent checks.

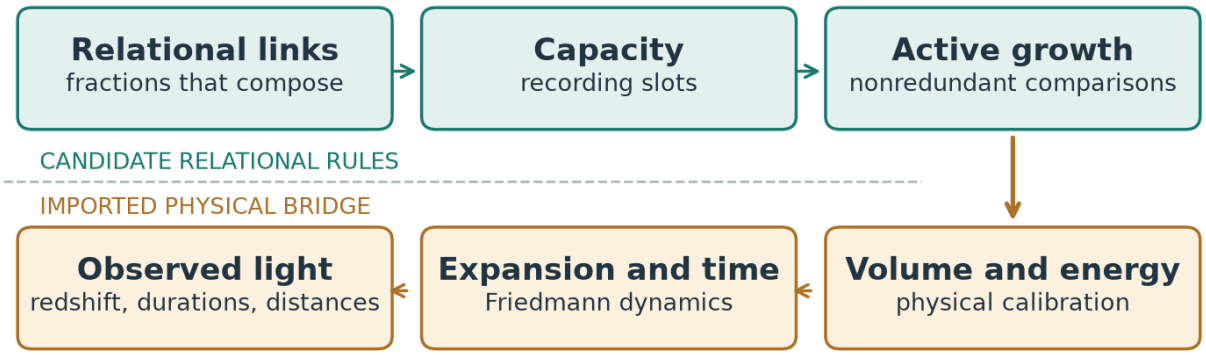


Figure 1. The path we will follow. The upper row contains the proposed relational rules. The lower row introduces their physical meaning and leads to observable signals. The change in colour marks a change in assumptions, not simply a change in vocabulary.

Ugo Lanciano | Ontological architecture by the author; mathematical development and writing with AI assistance. MM 2.32, integrated English revision 05, 16 September 2026. Independent article, not affiliated with a magazine or journal.

Why this may matter

Question	Answer
What is new?	An expansion history that follows from how a network of relations grows, with no function of redshift left free
What happened?	It held up against Pantheon+, DESI DR2 and the geometry of the microwave background, tested together
Did it beat the cosmological constant?	No. The dark sector carries three extra parameters, and the information criteria favour the cosmological constant
What is needed now?	Independent reproduction, and the full microwave-background likelihood in place of its compressed geometry

A distant galaxy reaches us through light that set out long ago. That light carries more than an image: it bears traces of its journey and of the expansion of the universe. We can read those traces, measure them and compare them with a theory. We know how to do this with considerable precision. What remains much less clear is the nature of what we describe as dark energy.

Mathematical Materiality, or MM, approaches the question from a different level. Before asking what substance fills space, it asks what makes objects, their distinctions and their relationships possible. Its proposed fundamental element is the **relational link**, a relation capable of connecting distinguishable states.

This article follows just one line of inquiry: can an organized growth of these relationships be translated into a component that accompanies cosmic expansion and can support its acceleration? The answer developed here is an explicit model, with an observational comparison and a clear boundary between counting and physics.

Independent cosmologists are invited to reproduce or falsify this result. The deposited materials hold the configuration, the scripts, the chain and the audit that checks them. A reproduction kit, which recomputes the headline numbers from public data with the code that produced them, will be deposited as a separate record that cites this one.

What the comparison establishes, and what it puts at risk

In short	What the comparison gives
Fit	The sampled chain reaches a total chi-square of 1420.02; the independently computed profile minimum is 1420.05. The difference is numerical, not physical. A cosmological constant in the same code, on the same data, reaches 1421.19.
Cost	Three additional dark-sector parameters. The one-unit improvement does not offset them, so model-selection criteria favour the cosmological constant.
Bound	The parameter p, which controls how far the model departs from a cosmological constant, lies below 0.0724 at 95 per cent credibility. This is an upper limit, not a measurement.
Falsifier	The growth law forbids the ratio w of the dark component from ever falling below -1; a measurement that establishes such a crossing would close the model.
Next test	The inhomogeneous sector, pre-registered in Appendix V with its decision rule fixed before the computation is run.

Appendices S to V give the full comparison. The three standard geometric probes are fitted at the same time, with the ordinary cosmological parameters left free: 1,590 Pantheon+ supernova light curves, thirteen baryon-oscillation measurements and three summary numbers for the geometry of last scattering.

The standard cosmological parameters come out of the fit rather than being imposed. The expansion rate and the matter density each land within about one standard deviation of the Planck values, within the standard framework for cosmic expansion and recombination that the analysis assumes.

A profile computed without the emulator reproduces the chain's minimum, and two further chains, not run to convergence, give consistent bounds. The two numbers that set when the departure switches on, and how sharply, remain free.

What follows is the construction itself: what a relational link is, how counting becomes an expansion history, and where the imported physics begins. A reader who wants the comparison first can go straight to Appendix S.

The question beneath the question

Imagine trying to explain a city. We can describe its buildings, streets and distribution of residents. We can also ask what relationships make that particular organization possible: which connections allow people to move, communicate and build something that lasts. These are two levels of explanation. The second does not erase the first, but changes our starting point.

MM attempts something similar for physical reality. The word *ontology* refers to this underlying question: which elements do we regard as primary, and which might result from their organization? In MM, objects and observers are conceived as configurations of the same relational fabric. The reference from which we describe something is part of the world being described.

This research grew around four questions. Why do physical laws retain their regularity across different places and epochs? What lies behind the phenomena attributed to dark matter? What mechanism might account for dark energy? And what picture of reality could make quantum behaviour intelligible?

Here we work only on the third question. Dark matter and dark energy have similar names but refer to different problems: the distal contribution studied in these pages concerns cosmological expansion, not an attempt to explain galactic rotation curves. We will not automatically transfer a result from one sector to another.

The image of a city helps distinguish levels of description; it is not physical evidence. Likewise, starting from relationships is not enough to obtain a theory. Rules are needed: we must specify which states are possible, how they change and what we can calculate. Only then can we check whether the idea leads to a history of the world consistent with observations.

Our aim is therefore concrete: to construct a path from the relational link to an observable expansion history, identifying every necessary step. The result will not rest on the evocative appeal of the word "relationship" alone.

A relational link, without mysterious language

We need only two registers to begin. We can imagine them as two boxes, each able to display a positive or a negative sign. They are small mathematical systems, not necessarily physical devices. "Recording" means only having a state distinguishable from another.

We connect the boxes with a link. We want their states to be neither completely unrelated nor forced always to agree. One simple approach assigns more internal possibilities to agreement than to disagreement.

Suppose, for example, that for a given first sign the link has three realizations that preserve it and one that reverses it. If all four possibilities are equally likely, the second register repeats the first three times out of four. Its excess of agreement over disagreement is the difference between three quarters and one quarter: one half.

We call this transmitted share of correlation q. In the example it is $1/2$. Notice the distinction: the probability that the signs agree is $3/4$, whereas the coefficient q is $1/2$. The coefficient measures how much the initial sign remains recognizable relative to a completely random choice, under the symmetric distribution adopted here.

If the counts were equal, the second register would carry no information about the first. If disagreement were impossible, the sign would pass through without attenuation. The model we will use lies between these extremes, with a positive fraction smaller than one.

No metres, seconds or masses have entered the construction. We have not introduced quantum entanglement either: this is a classical probabilistic channel. Its task is narrower and more precise: to give the "fractionating link" a mathematical realization based on finite counts.

At this level, the observer is not an external consciousness either. One register can be compared with another and serve as a reference. What it takes to construct a complex material observer is a further question; the mechanism described here does not require someone to be watching it in order to operate.

When a third register comes between two others

Now insert an intermediate box. The first link transmits half the correlation; the second transmits half again. One quarter remains at the end. The fractions multiply: half of a half is a quarter.

This is the operational meaning of *interposed fractionation*. The comparison between the endpoints passes through distinct steps, each contributing to the overall result. For two links we write $Q = q_1 q_2$; for a longer chain we multiply all the shares. The capital letter denotes the product over the entire path.

We can also turn this multiplication into addition by using a logarithm. The quantity $-\log Q$ increases as the retained correlation decreases; its individual contributions add along the chain. It measures the depth of mediation, still without physical units. The precise properties are proved in Appendix A.

What have we gained? A distinction between direct and mediated comparison that does not depend merely on the names assigned to objects. We have not yet obtained a spatial distance measured with a ruler.

An example clarifies the boundary. If I draw a connection as two segments rather than one, but choose their fractions so that the total product stays unchanged, nothing has changed. I have altered the drawing, not the system. It would be a mistake to make space grow every time we add a node to a graphical representation.

We therefore distinguish two operations: describing the same relation more finely, or generating a new state of the network. Only the second changes the available capacity. This precaution makes the mechanism independent of arbitrary detail in the drawing.

The product formula applies to the chain constructed here. In a network with multiple paths and cycles, correlations may receive different contributions: we will not replace that problem with a single-path formula without a proof.

Ten slots, eleven slots, twenty slots

Let us move from transmission to growth. We use n for the number of recording slots in the network. A simple rule replaces one slot with two at each update. The total therefore increases by one: ten, eleven, twelve, and so on.

To connect this growth with fractionation, we propose a specific channel between successive network states. When there are n slots, we assign $2n+1$ internal possibilities to agreement and one to disagreement. With ten slots, for example, the counts are twenty-one and one. Their difference divided by their sum is ten elevenths.

$$q_n = \frac{n}{n+1}, \qquad Q_{n_i,\,n_f} = \frac{n_i}{n_f}. \tag{1}$$

The second part of the formula needs no complicated calculation. From ten to twenty slots, we multiply ten elevenths, eleven twelfths, twelve thirteenths, all the way to nineteen twentieths. Every intermediate number cancels: ten twentieths, or one half, remains.

Capacity has doubled and the overall retained share has halved. In general:

$$C = \frac{n_f}{n_i} = Q^{-1}_{n_i,\,n_f}, \qquad u = \log C. \tag{2}$$

Here C is a growth factor, not a number of cubic metres; u is a convenient way to count that growth on a logarithmic scale, not yet a time measured by a clock.

[CANDIDATE MM RULE] We have chosen these counts to realize an exact connection between fractionation and capacity. The construction shows that the connection is possible and consistent; it does not prove that MM's ontology necessarily selects these counts. The dependence on the total number of slots is a collective rule whose strictly local origin remains to be explained.

It is also important to know where the chain lives: it connects successive states of the whole process under consideration. It does not say that every spatial path between two galaxies has the same fraction. The relationship between cosmological growth and observed distance will be constructed later.

A network can grow without preserving everything

Increasing the available slots does not mean increasing what remains active in the same proportion. Think of an archive in which some new comparisons add a usable distinction while others repeat something already reconstructed. The analogy separates total capacity from activity; it does not presuppose a cosmic archivist.

We represent this distinction with three labels: A and B denote two active types; I denotes a slot that is inactive for the count we are tracking. "Inactive" does not mean nonexistent: the slot still belongs to the overall capacity.

At each step, one slot is chosen at random, with every slot equally likely. If A is selected, it is replaced by A and B. If B is selected, two B slots usually appear; with a certain probability, A and I appear instead. If I is selected, two I slots appear.

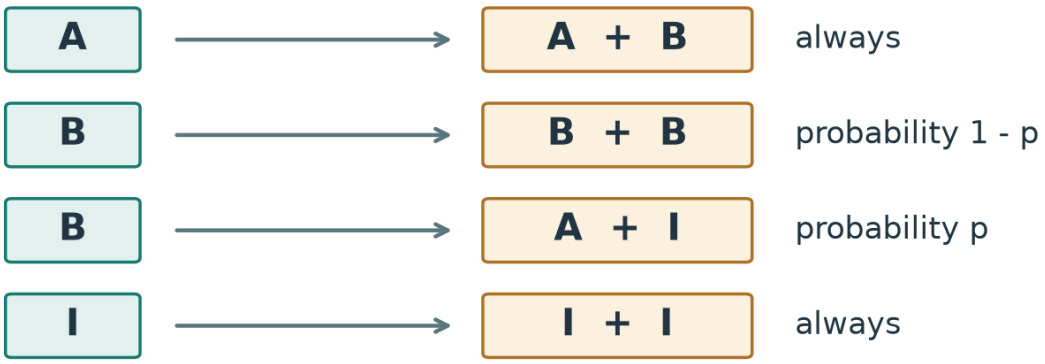


Figure 2. A proposed rule, not a process observed through a telescope. Each row replaces one slot with two. Active labels can increase or be redistributed; capacity always grows by one. The probability p concerns only the choice between the two outcomes for type B.

The letters A and B do not identify known particles. They are minimal states in a candidate mechanism. Their role is defined by the operations they permit, not by the names we give them.

An explicit rule lets us follow the process without deciding in advance which cosmological outcome we would like. Once fixed, it produces a distribution of possible histories. We can calculate their mean growth, fluctuations and response to changes in the rule itself.

Two probabilities that must not be confused

The possibility of obtaining A and I from B is called *redundancy*. We denote it by p. It has an elementary realization: draw two independent labels and regard the comparison as redundant when the labels coincide.

With four equally likely labels, there are sixteen possible pairs and four coincidences: the probability is one quarter. More generally, it is the sum of the squared label probabilities. This is an understandable statistical rule; the number four, however, is a choice in the illustrated model, not a cosmological result.

We now have two different quantities. The fraction q describes how much of a comparison is retained between successive states. The probability p governs which records remain active as capacity increases. Using the same word, "fraction", for both would hide an essential distinction.

The two constructions can coexist in a single probabilistic process. Appendix B gives the joint law and verifies that the probabilities sum to one. There is no need to pretend that either rule has automatically been deduced from the other.

Uniform selection is also a meaningful assumption. It says that a slot is not chosen more frequently merely because it carries the label A, B or I. Changing this could change the dynamics. We will use *robustness* to mean retaining a behaviour under specified changes, not under every imaginable modification.

This is an important difference between a free-form narrative and a model. In the former, we can give each word whatever behaviour the story requires. In the latter, we must accept the consequences of the stated rules, even when they do not lead to the desired outcome.

The average of possible histories

Repeat the game many times from the same initial state. The sequences will not be identical: A will be selected at one step, I at another, and chance will also affect B's outcomes. We can nevertheless ask how many active records there are, on average, after a given increase in capacity.

For this rule, the mean calculation closes exactly: the next step depends linearly on the current means. We do not have to invent an approximate form of the correlations to write that particular equation. The matrix in Appendix B is simply an orderly way to collect the contributions of the two active types.

When the initial number of slots is large, we can describe growth continuously. This means replacing many small updates with a curve while controlling the error in the specified limit. It does not mean that the network has thereby become a continuous spacetime.

In the long run, in the model's dominant regime, the mean active count grows as capacity raised to an exponent λ. That number can be calculated:

$$\lambda = \frac{1 - 2p + \sqrt{1 + 4p^2}}{2}. \tag{3}$$

There is no need to memorize the formula. Its message is this: once the redundancy rule is fixed, the relative growth rate is no longer free. For $p = 1/4$, it is approximately 0.809.

A power with an exponent smaller than one grows more slowly than its base quantity. Capacity can therefore keep increasing while the fraction occupied by active records decreases. At the same time, the total active count can grow. The two statements do not contradict each other.

The limit of the means is not enough, by itself, to describe a single realization. Appendix F therefore also retains the fluctuation calculation and specifies the intervals over which their relative impact decreases as the initial number of slots grows.

More activity, but less activity per slot

Consider a numerical example in the dominant regime, using the exponent just obtained. If capacity doubles, the active total is multiplied by about 1.75, not two. It has increased by roughly 75 per cent. Dividing by twice the capacity, however, reduces the active share per slot to about 88 per cent of its initial value.

This is the heart of the mechanism: **the total grows while its concentration becomes diluted slowly**. This is not yet a law of dark energy. It is the statistical behaviour that we will try to translate into physical language.

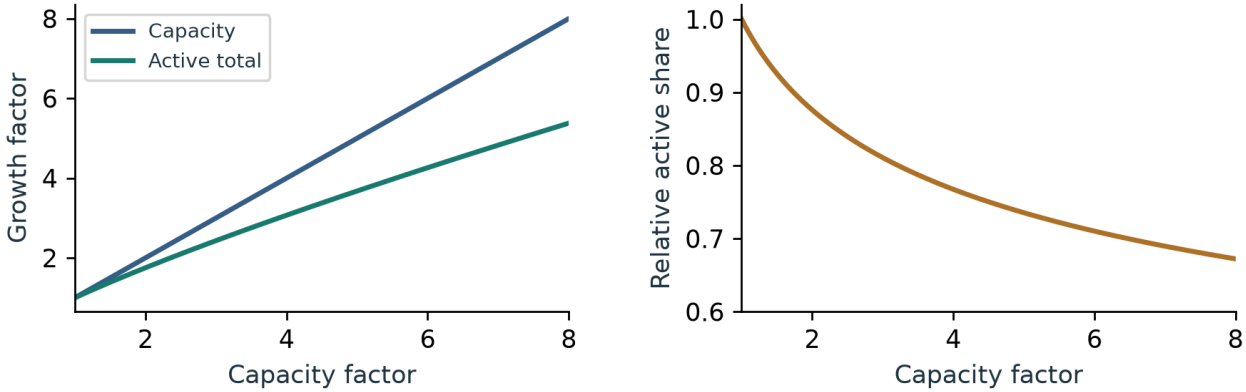


Figure 3. An illustration using only the dominant mode, with p=1/4. Capacity is compared with the normalized active total and their ratio. All curves start at one to make them easier to read: this does not mean that every initial slot is active. These are neither astronomical data nor the full curve used in the observational comparison.

The distinction is also clear if we imagine eight times the initial capacity. The active total grows by about 5.38 times; the ratio of active total to capacity falls to about 0.672. A quantity that was merely conserved, without new active growth, would instead be diluted eightfold.

Why might cosmology care? Because when a volume increases, what matters is not only how much energy it contains, but also how that amount changes during the increase. Two components with the same density today may have followed different histories and produce different evolutions.

Before speaking of energy, however, we must cross a boundary: a count of records does not automatically have the units of a volume or an amount of energy. That step deserves to be stated, not left for the reader to supply.

The dictionary that turns counts into physics

A map can be measured in centimetres on paper; to obtain kilometres, we need a scale. Here the issue goes deeper: we must not only choose a unit but decide which physical property a count represents.

We propose two identifications. The first assigns the same volume contribution to every slot: more slots mean more volume for the region considered. The second assigns the same energy contribution to each active record, using the mean active count. These are simple, testable rules. They are not proved by the fact that the link fractionates.

[IMPORTED EFFECTIVE POSTULATE] Volume is proportional to capacity and energy to the mean active count. Volume is interpreted within a cosmological geometry with three spatial dimensions. We adopt the work balance $dU = -P\,dV$ for the component considered, without additional exchanges with matter.

Let us read the last formula in words. U is the energy contained in the region; V is its volume; P is pressure. The letter d indicates a very small change. The minus sign specifies how energy changes when volume changes in this balance.

If energy increases during expansion, the pressure assigned by that balance is negative. If energy is proportional to volume raised to λ, the result is particularly simple:

$$w = \frac{P}{\rho} = -\lambda. \tag{4}$$

Here ρ denotes energy density: energy divided by volume. The number w compares pressure and energy density, which have the same physical dimensions. In the illustrative regime with $p = 1/4$, it is about -0.809.

The result has a precise meaning: a discrete rule selects a growth rate and, once this dictionary is adopted, that rate determines an effective pressure. We have not chosen the value of w directly; we have, however, chosen the microscopic rule and the physical dictionary. Both belong among the assumptions to be assessed.

Why negative pressure affects expansion

In everyday experience, we think of pressure as something that pushes against a wall. Cosmological pressure should not be imagined as a wind carrying galaxies outwards. It is a property of the component that enters the energy balance and the equations of gravity.

In general relativity, pressure also contributes to expansion dynamics. For a homogeneous component, the decisive term contains energy density plus three times pressure. If pressure is sufficiently negative, that contribution can support an acceleration of the scale factor. The threshold for a dominant component is $w < -1/3$ [2].

This is where we adopt the effective gravitational theory: counting records has not, by itself, proved Einstein's equations. The proposed derivation concerns a component to be inserted into those equations, with stated assumptions.

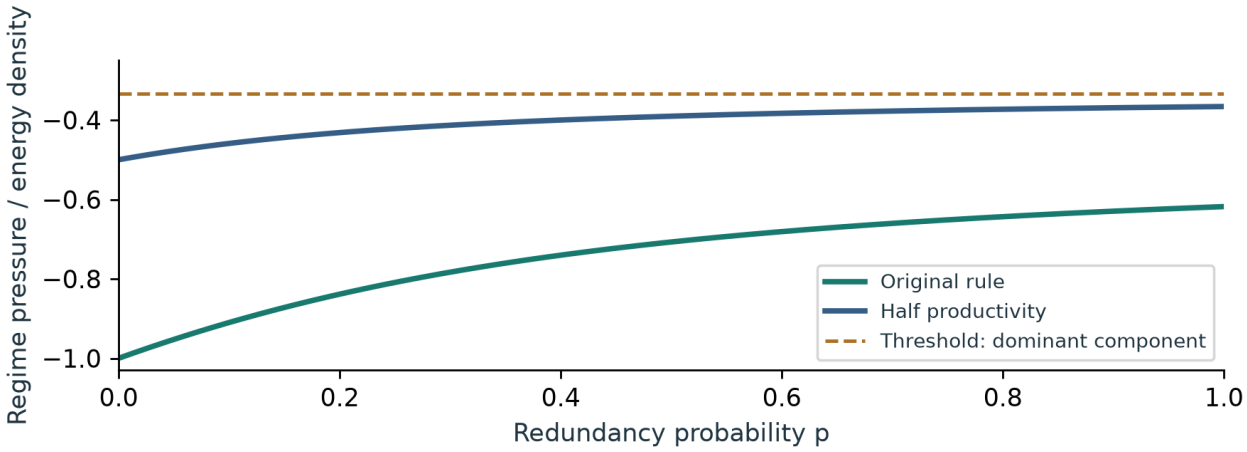


Figure 4. Regime values obtained from the rules, not measurements. The green curve uses the original branching rule. The blue curve halves the probabilities of producing new activity in the two channels specified in Appendix C. Both remain below the threshold line in the dominant case.

The blue curve answers a natural question: does the result disappear as soon as we touch a rule? Not in this family. We can reduce productivity and retain a range of accelerating behaviour. Appendix C identifies exactly which productivities guarantee that result for every redundancy probability considered.

It is not enough, however, for the distal component to cross the threshold in isolation. The model's universe also contains matter, which contributes to the dynamics. Overall acceleration depends on the balance among all components. The complete formula is retained in Appendix D.

How much space, in how much time?

If the number of slots represents a three-dimensional volume, eight times as many slots correspond to a factor of two in length: two cubed is eight. This introduces the *scale factor*, denoted by a, which describes changing separations between references that follow the mean expansion.

Between an epoch of emission and an epoch of observation, the adopted dictionary gives:

$$\frac{a_o}{a_e} = \left(\frac{n_o}{n_e}\right)^{1/3} = Q_{e,o}^{-1/3}. \tag{5}$$

The subscripts identify the epochs being compared: "e" for emission and "o" for observation. The cumulative link is thus connected to an increase in physical lengths through the identification of capacity with volume.

But doubling can take different amounts of time. Saying how much a quantity grows does not yet say how quickly it does so. Counting updates does not automatically provide a clock.

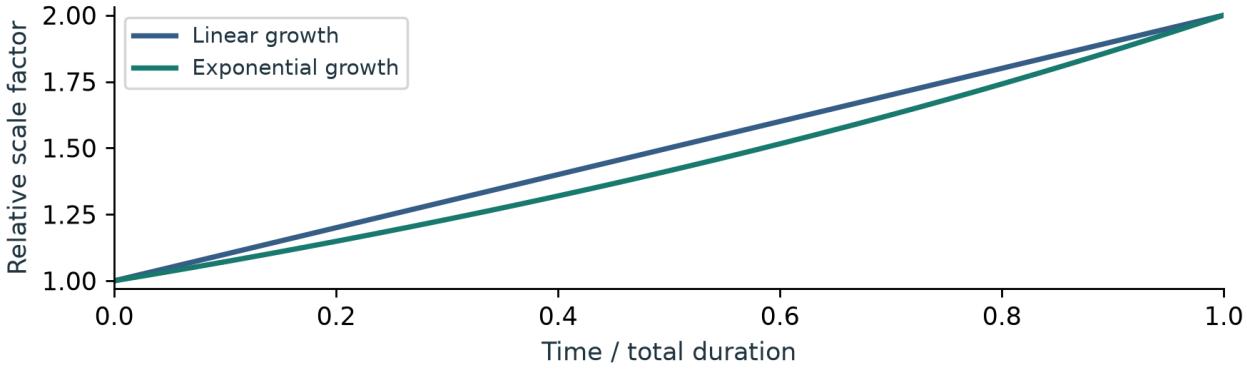


Figure 5. The same final growth and the same overall product of links, but different rates. One curve is linear; the other accelerates. This kinematic example shows that capacity alone does not determine the rate; it does not present two solutions of the same closed physical dynamics.

To obtain that rate, we adopt cosmological proper time, meaning the time of ideal observers who follow the mean expansion, together with the Friedmann equations. Combined with the energy balance, they close the homogeneous system: given the initial data, we can calculate how the history unfolds.

The clock is therefore connected consistently to the effective dynamics, not derived from the algebra of the link alone. This distinction lets us reach a physical prediction without crediting the network with a deduction we have not made.

"Distal" means comparing epochs

Looking far away means receiving light emitted in the past. Observed distance and the history of the universe therefore meet in the same signal. In this model, *distal* denotes the accumulation of interpositions along the succession of states separating two epochs being compared.

It does not mean that a region must have a higher density today simply because it is far from us. Another observer following the same mean expansion does not become the physical centre of a different universe. The comparison changes its reference; the homogeneous cosmological law must remain shared.

Light provides a consistency check on the bridge. In the adopted geometry, wavelengths stretch with the scale factor. We call the relative increase in wavelength *redshift*: if the wavelength doubles, the redshift is one, not two. The stretching factor is $1+z$.

Two light-wave crests emitted a short time apart also arrive farther apart in time. In the limit of very small intervals, the same relation holds for durations:

$$1+z=\frac{d\tau_o}{d\tau_e}=Q_{e,o}^{-1/3}. \tag{6}$$

In words: redshift, the stretching of durations and growth between the two epochs are described by the same factor. For finite intervals, the approximation requires them to be short compared with cosmological variation. This is neither a new explanation of quantum behaviour nor a derivation of local relativity from the binary channel.

[IMPORTED EFFECTIVE POSTULATE] Light propagation and the meaning of clocks are those of standard cosmological geometry. Photon number is conserved in the bridge used for luminosity distance.

This last qualification avoids counting an effect twice. We must not multiply luminosity by Q again as though it were a photon's survival probability. The fraction belongs to the relational register; light is already described by the adopted propagation law [1].

What we ask of supernovae

A cosmological theory must encounter measurements, not just suggestive images. Here we use an already completed comparison with Type Ia supernovae: explosions whose light curves, after appropriate corrections, provide distance indicators. They are not identical light bulbs; standardization and uncertainties are part of the analysis [3].

The model supplies an expansion history. From that history we calculate the relationship between redshift and apparent brightness. The restricted question is: does a curve fixed before the new calculation remain comparable with another set of observations, without changing its physical shape to fit the new points?

The first data set used to determine the parameters contained Pantheon+ data and seven DESI BAO observables in the domain up to redshift one. "BAO" denotes a characteristic scale in the cosmic matter distribution, used as a distance reference. That work is described in Section B18 of the MM 2.30 volume [11, 14].

The subsequent comparison used the DES-Dovekie reanalysis. Objects already associated with the comparison inventory, and two ambiguous associations, were excluded. This leaves 1,448 distinct DES objects under the documented criteria. The original Pantheon+ unique-object count was also 1,448: a numerical coincidence between different sets, not the same sample renamed.

The physical parameters were retained at their earlier values. Only a uniform shift of the brightness scale was refitted on the new observations, as required when absolute calibration is not fixed. Moving an entire curve vertically is not the same as changing its shape.

The protocol was fixed before acquiring and calculating the new individual distances, but after the published aggregate DES results were known. **This was neither a blinded test nor an external preregistration.** Catalogues without shared objects may also share calibrations. These conditions limit the strength of the comparison and remain part of the result, not notes to be discovered afterwards.

The curves meet the observations

How do we summarize a comparison involving more than a thousand measurements? One quantity used is called *chi-square*. It combines the differences between curve and data, accounting for uncertainties and their correlations. In the comparison reported here, a smaller value means smaller overall discrepancies under the same procedure.

Model transferred to the DES data	Chi-square
Cosmological-constant reference	1263.358
Constant-w control	1264.875
MM B17 distal component	1265.041

Table 1. Results from the earlier local study, not recalculated as a new test for this edition. Each model retains its own matter parameter calibrated in B18. The constant-w control also retains the vacuum residue specified in Appendix G. Differences are not converted into probabilities that a theory is true.

The three curves are very close. MM is not preferred: its chi-square exceeds that of the cosmological-constant reference by about 1.68. The comparison with the constant-w control also shows how little these data distinguish the distal mechanism from another expansion history.

This leaves a restricted result intact: the model reaches an observational curve that, in this relative comparison, does not depart sharply from the adopted references. It does not, however, provide an exclusive validation of relational links as the origin of expansion.

There is a further statistical caution. The absolute chi-square values are low relative to a simple Gaussian reading. The DES procedure incorporates weights and corrections: we cannot directly convert these values into a reliable percentage of theoretical validity. A discrepancy in the published effective count remains to be clarified. Appendix H reports the numbers without hiding them or giving them a meaning they do not have.

We will therefore not use "test passed" as a stamp of approval. We will state exactly what happened: transfer of the curve with frozen physical parameters, relative comparison close to the controls, no preference for MM and no available absolute validation probability.

How far this story reaches

The observational comparison is restricted to redshift no greater than one. In scale-factor terms, this follows the phase in which physical distances between comoving references have grown up to twice their value at the chosen past endpoint. This solution is not a reconstruction of the entire history of the universe.

The boundary is not merely editorial. If we extend the frozen solution backwards, the mean type-B count reaches zero near $z = 1.665$ and would then become negative. A negative number of records is not an admissible state. That continuation must therefore stop or be replaced by a further physical bridge. Appendix H documents the check.

Within the analysed interval, the solution remains admissible. The same history cannot, however, be used unchanged for the primordial universe. In particular, we do not present an MM comparison with the cosmic microwave background or with all high-redshift BAO measurements.

The model nevertheless has consequences that can already be challenged within its domain. For example, with the stated rules and dictionary, the distal component's pressure-to-density ratio does not fall below -1. In the frozen curve it is about -0.815 today and -0.875 at redshift one. The regime value illustrated earlier, about -0.809, is a limit of the dynamics: it must not be confused with the present value of the full curve.

A robust, directly comparable observational reconstruction requiring behaviour incompatible with these bounds would challenge this version. Preferences for varying dark energy discussed in DESI analyses depend on the data combination and parameterization: they are a comparison to address, not an automatic refutation inferred from a result's headline alone [9].

Being able to revise a theory does not authorize changing its rules after every result without recording the change. An extension beyond the current domain would be a new or enlarged model, with its own assumptions and checks. Here we deliver the restricted contribution we have actually constructed.

A proposal within a scientific conversation

The idea that geometry might have a more fundamental description did not begin with MM. Recognizing neighbouring work helps identify which question is shared and which construction differs. It does not assign to MM results obtained elsewhere.

In **causal sets**, spacetime is described through discrete elements and an order representing a causal relation. Element counts are related to spacetime volume [4]. This has an affinity with moving from counts to geometry, but the spatial volume of a region at one epoch in our model is not their spacetime volume. Our binary channel does not already contain that causal order.

In **Everpresent Lambda**, Ahmed, Dodelson, Greene and Sorkin study a fluctuating cosmological constant motivated by unimodular gravity and causal sets [5]. Here we instead follow active growth and dilution of a component, keeping a possible constant vacuum residue separate. Fluctuations in our counts do not automatically inherit that programme's prediction.

Padmanabhan proposed interpreting expansion through an imbalance between degrees of freedom associated with a surface and its interior [6]. The connection lies in linking expansion with counts. The mechanism differs: MM does not calculate those two numbers here or derive their balance law.

Oriti, Sindoni and Wilson-Ewing derive cosmological dynamics from particular collective states in group field theory, also using a scalar field as a time reference [7]. This is another route from a microscopic description to cosmology, with quantum structures of its own. Our statistical model does not replace them through a simple analogy.

Finally, the mathematics of branching belongs to a long-studied family: **generalized Pólya urns** and multitype processes, including Janson's work [8]. Drawing an element and changing the composition according to type-dependent rules is not a formalism invented by MM.

What sets this construction apart is where it places the primitive: not a finer piece of geometry, as in most of these programmes, but a relation, the possibility of telling two states apart. Space does not contain such differences; it counts them.

The proposed contribution is the specific chain connecting the fractionating channel, capacity, active growth and a cosmological dictionary. Whether and to what extent this chain is original requires a broader literature comparison: we do not claim general priority for the emergence of space from degrees of freedom.

How a network can grow without stretching every structure

The capacity model has brought us to an expansion history. A further question now becomes concrete: must every new record enlarge space? If it did, how could a structure retain its size while the surrounding network grew?

Imagine a survey of a changing town. Some entries confirm a connection already described. Others establish a new intermediate location that changes the route between existing places. Both add to the archive, but only the second changes the town's spatial description. MM can express that distinction as a candidate rule, rather than leave it as an analogy.

[CANDIDATE MM RULE] Give each spatial register a symbol and read the endpoints of a link through a noisy channel. Equal readings create a memory record. Different readings request an interposition: a new spatial register between the endpoints. The request is realized only when the neighbouring cells can be updated compatibly. Otherwise it remains a memory of an unrealized distinction.

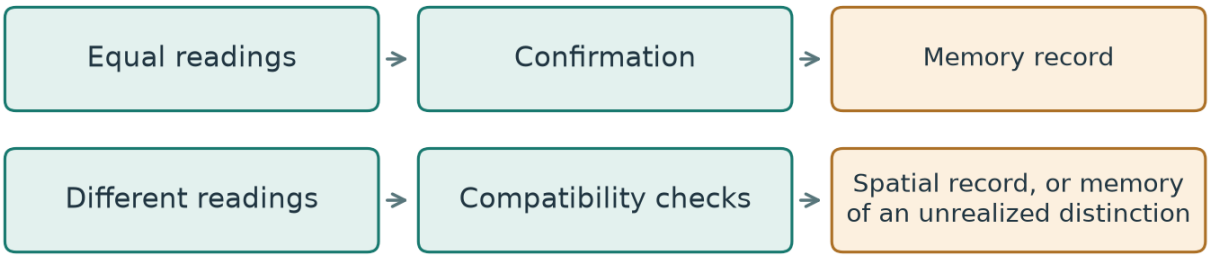


Figure 6. Two kinds of record, not two kinds of substance. Compatibility can require further comparisons, each with its own classified outcome. The diagram describes a candidate update rule; it is not an observation of microscopic space.

This is a new exploratory model [16], not a relabelling of B17. It distinguishes spatial registers, memory records and the count to which energy might be assigned. The supernova comparison in Table 1 still belongs exclusively to the original B17 construction.

The gain is a more explicit question: which relational changes alter the current spatial organization, and which preserve a record without doing so? Appendix I defines the rule; the remaining appendices show what can already be calculated from it.

A stable island in a growing network

Consider a region whose registers carry the same symbol and whose internal channels transmit perfectly. Every internal reading then confirms agreement. Under the proposed rule, these comparisons add memories but do not subdivide the region's internal spatial links.

Outside that region, imperfect channels can produce different readings. Compatible interpositions increase separations measured along the surrounding spatial graph. The same update rule can therefore accommodate a persistent organization and a growing environment. It does not need a separate command saying that this particular object must be held fixed.

There is numerical evidence for this mechanism in the exploratory mesh model. In the recorded run with a coherent region, its mean internal graph distance remains about 1.32 through a total of 131,072 records. In a control with imperfect channels in that region, the internal distance instead grows to about 6.96 [16]. These are graph steps, not metres, and the run is an internal model experiment, not an astronomical measurement.

What is supplied initially matters. The coherent states and perfect internal channels are given to the model. It demonstrates persistence for that preparation; it does not explain how ordinary matter first forms such an organization. Three-dimensional cell structure is also supplied, not selected by the dynamics from an amorphous network.

The model separates two descriptions that should not be conflated. The cells can be refined inside a reference domain whose coordinate volume stays fixed, while paths measured in current graph steps become longer. Interpreting that second change as physical expansion requires a physical calibration. A denser mesh alone is not a proof that the universe has expanded.

Even with that boundary, something useful has been gained: expansion need not mean identical behaviour for every relational organization. Stability and growth can be addressed within one explicit candidate mechanism. Formation, physical length calibration and gravitational binding remain the next bridges, rather than unspoken ingredients of the demonstration.

Counting the work inside a comparison

A new interposition may require its neighbours to adjust first. Think of replacing a junction: neighbouring routes must still meet consistently after the change. In the mesh, this creates a chain of further comparisons. A requested change can fail, yet successful earlier changes remain legitimate events because each has its own reading and record.

Counting only the initial request misses this internal work. The useful quantities are the average numbers of spatial records and memories produced by an entire cell event, including the comparisons it requires. We denote them by the growth coefficients gamma-S and gamma-memory.

An idealized chain model makes these averages calculable. It assumes independently sampled requirements and a fixed confirmation probability. Its closed equations remain useful, but a network remembers: two links with the same number of records can carry different states, and therefore have different chances of producing a new spatial record.

The next step is now explicit. Keep the states, the current cells and the order of compatibility checks. The reading rule then determines the probabilities of the possible event histories. Summing their spatial and memory outputs defines the growth coefficients. Appendix O gives this state-dependent formulation. No average confirmation probability has to be inserted into the microscopic evolution.

Eight new homogeneous-network runs follow this rule through completed refinement levels. With a memoryless channel, their mean yields are about 0.928 spatial records and 0.432 memories per cell event. With a channel that retains half of each endpoint's symbol signal, they become about 0.893 and 0.458. The measured confirmation share rises from about 0.249 to 0.266. These are finite-network experiments, not astronomical observations.

The difference has a precise meaning: retaining information changes growth even when the rule for recording is unchanged. Appendix P proves that the total numbers of spatial and memory records alone cannot determine this evolution. Appendix Q describes the complete cohorts, the independent random seeds and the remaining finite-size limitation. Historical short-run comparisons remain identified as such in Appendix L.

The independent-chain formula still approximates the new results when supplied with their measured statistics; it is not thereby an autonomous prediction. A newer pair approximation improves some geometric averages but fails a consistency check, documented in Appendix R. Neither approximation is needed to execute the state-dependent rule. This is the completed, restricted advance: a specified candidate process with reproducible full-level tests, not a unique microscopic theory extracted from MM's ontology.

The energy choice becomes visible

We can now revisit the physical dictionary with more structure. The original B17 comparison assigns volume to total capacity and energy to the active count. The exploratory mesh distinguishes spatial records from memories, so it offers a different candidate dictionary: volume proportional to the spatial count, with energy assigned to a specified subset of records.

If every spatial record carries the same effective energy and the same volume contribution, their ratio is constant. With the separately adopted work balance, the corresponding component has pressure-to-density ratio minus one. This is a possible microscopic interpretation of a constant-density component, not a new fit or a replacement for B17's changing active fraction.

[IMPORTED EFFECTIVE POSTULATE] Assign physical volume and energy to the selected counts, and use the no-exchange work balance for the component being modelled. The general relation between their growth rates is given in Appendix N. Neither the energy unit nor the volume unit is determined by the counting rule alone.

It is possible instead to assign energy to unrealized distinctions as well. Their accumulation near a coherent region then becomes relevant. The exploratory runs show a strong enhancement of this count near the region, compared with a control; they do not yet establish a physical energy profile or its gravitational field. A choice about energy must not be hidden inside the word "memory".

General relativity does not require this choice to favour spatial records. Its source includes energy, momentum and stresses; failing to create extra volume does not by itself make a record energetically irrelevant [15]. Assigning energy only to spatial records is a minimal modelling option within MM, not a deduction imported from Einstein's equations.

This extension therefore sharpens the problem rather than changing the observational result already reported. It supplies a distinction between memory and spatial organization, a persistence mechanism and a calculable approximation to growth. It also exposes the remaining physical decision: which distinctions carry energy, with what coupling and calibration?

The answer can be revised as the microscopic model improves. For the present edition, the cosmological equations, fitted parameters and supernova comparisons of B17 remain unchanged. The new mechanism is an explicit research extension alongside that tested effective construction.

From links to an expansion history

We began by asking whether relationships could be more fundamental than objects and space. The resulting operations can now be reconstructed and discussed.

The construction connects integer-count channels, cumulative fractionation, growing capacity and active comparisons. A stated physical dictionary then carries calculable growth to density, pressure, expansion and light signals.

We can identify where each consequence comes from, change a premise and calculate how much the result depends on it.

Its value does not require observations to be new or different from those described by other theories. Observations are what every description must confront. An alternative theoretical organization can be interesting if it makes explicit connections previously left separate. But agreement with a known trend does not, by itself, identify the physical origin of that trend.

The memory-space extension adds a distinction between recording and interposition, persistence of a prepared coherent region, and conditional growth equations. It now specifies the state-dependent reading process, proves why counts alone cannot close it exactly for all states, and tests eight completed refinement cohorts. These advances do not select B17's dynamics or inherit its observational comparison as their own validation.

Open questions include the local origin of the collective rule, formation of coherent regions, deriving space and clocks from relational premises alone, calibrating energy and volume, and connecting fluctuations with observable perturbations. The value of the vacuum residue is unexplained. These are limits of the construction delivered here, not additional promises included in the result.

The restricted thesis is this: an explicit relational model exists that, through a stated physical dictionary, generates a component capable of supporting accelerated expansion and reaches a limited observational comparison. The comparison does not favour it over alternatives. The construction remains assessable as a conditional mathematical and physical proposal.

For readers who want to check the steps

Appendices A-F retain the original derivations; G-H specify data and domain. I-N develop memory, space, chains and energy alternatives. O-R give the state-dependent law, the obstruction to count-only closure, completed-level tests and the pair audit. S-V carry the comparison with supernovae, baryon acoustic oscillations and the compressed microwave background, what it bounds, where the treatment of the local noise stops being valid, and the test fixed in advance for the sector beyond it. Both reading paths stand without the conversation that produced them.

General readers may stop here, or continue directly to Appendix S for the new observational result. Appendices A-R contain the derivations and the model-development record.

A | The link, with a well-defined probability

[CANDIDATE MM RULE] A channel connects two registers $\sigma, \sigma' \in \{-1, +1\}$. There are a equally likely internal realizations for each agreeing pair and b for each disagreeing pair, with integers $a > b \geq 1$. Normalizing the rows of the count matrix gives:

$$T = a\mathsf{I} + b\mathsf{J}, \quad P(q) = \frac{T}{a+b}, \quad q = \frac{a-b}{a+b}. \tag{A1}$$

Here I is the identity and $\mathsf{J}(x,y)^T = (y,x)^T$ swaps the components. We adopt the uniform sign distribution, which is stationary. Moreover, $\mathbb{E}[\sigma' \mid \sigma] = q\sigma$. Iterating this identity along an open chain gives $\mathbb{E}[\sigma_i \sigma_f] = Q = \prod q_e$. The matrices share the same two eigenvectors, so $P(q_1)P(q_2) = P(q_1 q_2)$. A tree has a unique path; on a graph with cycles, this formula does not generally hold for the correlations of the joint distribution.

[DEDUCTION WITHIN THE MODEL] On unique paths, $d_l = -\log Q$ is a dimensionless additive length, positive if every edge has $q < 1$. It is not a Lorentzian metric and does not determine the dimension of space.

If relative capacity is a positive continuous function $F(Q)$, with $F(xy) = F(x)F(y)$ and growth as Q decreases, set $g(s) = \log F(e^{-s})$. Continuity and additivity of g on $s \geq 0$ imply $g(s) = \alpha s$, with $\alpha > 0$ in the nonconstant case. Thus $F(Q) = Q^{-\alpha}$. The form follows from the functional assumptions; the identification with capacity and the value of α do not.

The instance adopted here chooses $\alpha = 1$, $a_n = 2n + 1$, $b_n = 1$. Then:

$$\prod_{n=n_i}^{n_f-1} P(q_n) = P\left(\frac{n_i}{n_f}\right), \quad \sum_{n=n_i}^{n_f-1} (-\log q_n) = \log \frac{n_f}{n_i}. \tag{A2}$$

The case $\alpha = 1$ is a bridge assumption, not an experimental determination or an ontological theorem. The sequence lives on the graph of the process's epochs, not on every spatial branch. The internal channel counts are neither the slot count n nor the active count K. The earlier multiplicity descriptor W is not identified here with any of the three.

A representational subdivision preserving the product preserves A2. An update $n \mapsto n+1$ changes that product: growth therefore does not depend on an arbitrary number of segments in a drawing.

B | One process for capacity and records

The state is (A, B, I, σ), with nonnegative integer counts, $A + B + I = n \geq 2$. A slot is chosen uniformly and replaced according to the narrative's rules. Let $\mathcal{K}_n(x, x')$ be the transition kernel for the counts $x = (A, B, I)$. Its probabilities sum to one and every successor state has total $n + 1$.

[CANDIDATE MM RULE] The binary history channel and branching are conditionally independent given the state and n. The joint kernel is therefore:

$$\Pr(x', \sigma' \mid x, \sigma) = \mathcal{K}_n(x, x') P(q_n)_{\sigma\sigma'}. \tag{B1}$$

B1 is normalized and nonnegative. It realizes both product A2 and the active dynamics, without claiming that one imposes the other. Global capacity appears in the law: its generation by a local causal mechanism remains to be obtained.

For $X = (A, B)^T$, the possible increments are $v_1 = (0, 1)^T$, $v_2 = (1, -1)^T$ and zero. Their probabilities are $[A + (1 - p)B]/n$, pB/n and I/n, respectively. Consequently:

$$\mathbb{E}[\Delta X \mid X] = \frac{MX}{n}, \quad M(A, B)^T = (pB, A + (1 - 2p)B)^T. \tag{B2}$$

Setting $m_n = \mathbb{E}X_n$, we obtain without approximation:

$$m_{n+1} = \left(\mathrm{I} + \frac{M}{n}\right) m_n. \tag{B3}$$

Closure is exact because the mean increment is linear and the total n is deterministic. For an eigenvalue λ of M, the multiplier from step n_i to step n_f is:

$$\prod_{n=n_i}^{n_f - 1} \left(1 + \frac{\lambda}{n}\right) = \frac{\Gamma(n_f + \lambda)\Gamma(n_i)}{\Gamma(n_f)\Gamma(n_i + \lambda)}. \tag{B4}$$

As $n_i \to \infty$ at fixed ratio n_f/n_i, B4 tends to $(n_f/n_i)^\lambda$ with relative error $O(1/n_i)$. This gives $dm/du = Mm$, with $u = \log(n/n_i)$. It is a controlled limit of the means; approximating a single realization also requires the fluctuation control in F.

C | Why active growth is robust

The eigenvalues of B2 are those of Equation 3 with both signs in front of the square root. For $0 < p \leq 1$ and nonzero initial activity, the dominant mode is accessible and positive. In the continuous limit, write $\bar{K} = m_A + m_B$ and $f = m_B/\bar{K}$. We obtain:

$$\frac{d\log \bar{K}}{du} = 1 - pf, \qquad \frac{df}{du} = 1 - (1+2p)f + pf^2. \tag{C1}$$

Here f is the ratio of means, not the mean of the random ratio. The interval $[0,1]$ is invariant: the vector field equals 1 at zero and $-p$ at one. For two solutions f, g with the same p, the difference of their vector fields is $-(f-g)[1+p(2-f-g)]$. Their separation therefore does not exceed $e^{-u}|f(0)-g(0)|$.

The stable fixed point is $f_* = (1+2p-\sqrt{1+4p^2})/(2p)$ and $1 - pf_* = \lambda_+$. The transient is determined by the same initial data; it is not removed before the observational comparison.

A broader family reduces productivity: A produces A+B with probability s, otherwise A+I; B produces A+I with probability p, B+B with probability $(1-p)t$, otherwise B+I. All transition probabilities remain nonnegative for $s, t \in [0,1]$.

$$M_{s,t}(A,B)^T = (pB, sA + bB)^T, \quad b = t(1-p) - p. \tag{C2}$$

$$\lambda_+ = \frac{b + \sqrt{b^2 + 4sp}}{2}, \quad \lambda_+ > \frac{1}{3} \iff t + p(3s - t - 1) > \frac{1}{3}. \tag{C3}$$

The second root is nonpositive. Evaluating the characteristic polynomial $x^2 - bx - sp$ at $x = 1/3$ proves C3. The inequality has a strictly positive margin for every $p \in [0,1]$ exactly when $s > 4/9$ and $t > 1/3$. This is a result about the model, not the measured parameters of the universe.

For $p = 0$, the redundant channel disappears; the original rule gives $\lambda_+ = 1$. This is a formal limit: finitely many normalized label probabilities cannot give $\sum \pi_j^2 = 0$. Nor does the limit automatically eliminate fluctuations.

D | The physical bridge and the closed clock

[IMPORTED EFFECTIVE POSTULATE] Consider a homogeneous comoving domain: $V = v_* n$, $U_D = E_* \bar{K}$, with positive constants. In three dimensions, $a/a_i = (n/n_i)^{1/3}$. We use energy density $\rho_D = U_D/V$ and pressure $P_D = -dU_D/dV$, without additional exchanges with matter. This gives:

$$w_D = -\frac{d\log \bar{K}}{du} = -1 + pf, \qquad \frac{d\rho_D}{du} = -(1 + w_D)\rho_D. \tag{D1}$$

In the general case C2, $w_D = -s(1-f) - t(1-p)f$. Positivity and $0 \le s, t, p, f \le 1$ give $-1 \le w_D \le 0$. On the dominant mode, $\rho_D \propto a^{-3(1-\lambda_+)}$ and $w_D = -\lambda_+$.

[IMPORTED EFFECTIVE POSTULATE] We adopt cosmological proper time τ, spatially flat FLRW geometry, general relativity, separately conserved nonrelativistic matter and a possible constant residue ρ_v. Radiation is neglected in the late-time domain considered. With c explicit, a closed system is:

$$H^2 = \frac{8\pi G}{3c^2}(\rho_m + \rho_D + \rho_v), \quad \frac{du}{d\tau} = 3H, \quad \frac{dm}{d\tau} = 3HMm. \tag{D2}$$

Choose the branch $H > 0$, positive total density and initial data; also $\rho_m = \rho_{m,i} e^{-u}$. Differentiating the Friedmann constraint, D1 and D2 imply:

$$\dot{H} = -\frac{4\pi G}{c^2}[\rho_m + (1 + w_D)\rho_D]. \tag{D3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}[\rho_m + (1 + 3w_D)\rho_D - 2\rho_v]. \tag{D4}$$

Dots denote derivatives with respect to τ, which also defines the durations. Consistency among the constraint, continuity and acceleration can therefore be checked: no growth-rate law incompatible with the energy balance is added.

If only the distal mode dominates, then for $\lambda_+ < 1$, $a \propto (\tau - \tau_*)^{2/[3(1-\lambda_+)]}$; for $\lambda_+ = 1$, $a \propto e^{H\tau}$. These are regime solutions, not a reconstruction of the Big Bang or the actual age of the universe. With matter present, D4 is needed; checking $w_D < -1/3$ alone is insufficient.

E | From the universe's rate to light signals

[IMPORTED EFFECTIVE POSTULATE] Light follows null geodesics of the FLRW metric and photon number is conserved. For a comoving emitter and receiver, a radial path satisfies:

$\chi = \int_{\tau_e}^{\tau_o} c\, d\tau / a(\tau)$.

Two nearby crests traverse the same comoving interval. Subtracting the two integrals gives:

$$\frac{d\tau_o}{a_o} = \frac{d\tau_e}{a_e}, \qquad 1 + z = \frac{a_o}{a_e} = Q_{e,o}^{-1/3}. \tag{E1}$$

This is a cosmological relation between events, not a local Lorentzian dilation derived from register algebra. Setting today's $a_o = 1$ and $\mathcal{E}(z) = H(z)/H_0$, in the flat case:

$$D_L(z) = (1+z)\frac{c}{H_0}\int_0^z \frac{d\zeta}{\mathcal{E}(\zeta)}. \tag{E2}$$

For comparison with supernovae, the distance modulus is $\mu = 5\log_{10}[D_L/(10\,\mathrm{pc})]$, with the usual calibrations and an offset when absolute luminosity is not fixed. E1 and E2 are the standard bridge [1], not a new law of light propagation.

To calculate $\mathcal{E}$ without using only the asymptote, set $a_i = 1/2$, $p = 1/4$, $f_i = 1/2$. The initial normalization of m_i cancels in the ratio:

$$D(a) = \left(\frac{1}{a}\right)^3 \frac{\mathbf{1}^T e^{3M\log(a/a_i)} m_i}{\mathbf{1}^T e^{3M\log(1/a_i)} m_i}. \tag{E3}$$

$$\mathcal{E}^2(a) = \Omega_m a^{-3} + \Omega_v + (1 - \Omega_m - \Omega_v) D(a). \tag{E4}$$

Here $\mathbf{1}^T = (1, 1)$ and $D(1) = 1$. The inherited comparison uses $\Omega_v = 0.02$ and $\Omega_m = 0.2767981374$ for B17; the cosmological-constant reference uses $\Omega_m = 0.3272014632$. Parameters and normalizations come from the local study, not from the links alone. The transfer domain is $0 \le z \le 1$.

The new realization A2 does not change E3-E4: it completes an interpretive connection compatible with the same capacity history. This article therefore claims neither an improved fit, a new data selection nor observational validation of A2. An unknown cross-covariance between calibration and validation is treated as zero in the inherited comparison, not proved absent.

F | Fluctuations, calibrations and alternatives

The discrete model lets us track fluctuations before giving them an astronomical meaning. Let $\Sigma_n = \mathrm{Cov}(X_n)$ and, for the original rule, $D(m) = [m_A + (1-p)m_B]v_1 v_1^T + p m_B v_2 v_2^T$. The conditional second moment of the increments is $D(X)/n$. Applying the second-moment balance gives exactly:

$$\Sigma_{n+1} = \Sigma_n + \frac{M\Sigma_n + \Sigma_n M^T + D(m_n)}{n} - \frac{(Mm_n)(Mm_n)^T}{n^2}. \tag{F1}$$

For deterministic initialization and a fixed ratio bound $n/n_i \le R$, the increments are bounded. F1 gives a bound $S_{n+1} \le (1 + C/n)S_n + C'$ for $S_n = \mathrm{tr}\,\Sigma_n$, with constants depending on R and the rule. Iteration gives $S_n = O(n_i)$; hence $\mathrm{Var}(K_n/n) = O(1/n_i)$. Chebyshev's inequality justifies the ratio limit on finite relative-capacity intervals. It does not justify a uniform limit for all future times at fixed n_i.

Translating F1 into a spatial spectrum, gravitational lensing or anisotropies requires correlations between regions, causal transport and a coarse-graining law. **This observational signature has not yet been derived.** Separating counts does not determine a physical sound speed; the parameter w cannot simply be assumed to equal its square.

The importance of the physical dictionary becomes clear when it is changed: if $V \propto n^\nu$ and energy per register grows as n^η, the dominant mode gives $w = -(\lambda_+ + \eta)/\nu$. The prediction therefore also depends on structural calibrations, not just the matrix. Adding $U_v = \rho_v V$ leaves the counts unchanged but changes the cosmology: the vacuum residue remains separate and unexplained.

Finally, a homogeneous history with $\rho_D > 0$ and $-1 < w_D < 1$ can be reproduced by a canonical scalar field through $\dot{\phi}^2 = (1 + w_D)\rho_D$ and $V_\phi = (1 - w_D)\rho_D/2$, in units $c = 1$ [2]. This is an alternative effective completion, not a microscopic derivation of that field from MM. It shows why agreement on distances does not uniquely identify a relational origin.

G | Reading and reconstructing the comparison

Table 1 reproduces the local transfer calculation [14], not a new fit. A separate and wider comparison, which does refit, is reported in Appendices S-U and does not modify this table. The parameters come from B18: 1,565 Pantheon+ light curves, 1,448 distinct objects and seven BAO observables within redshift one.

Model	Frozen matter parameter
B17	0.2767981374
Cosmological-constant reference	0.3272014632
Constant-w control	0.2713974385

The control retains $w = -0.8151435616$ for its nonconstant component and $\Omega_v = 0.02$, as in B17. It is therefore not accurately described, without qualification, as a model containing a single constant-w dark component. In the cosmological-constant reference, the entire nonmatter component is constant.

The DES selection retains survey code 10 and $0.10 \le z_{HD} \le 1$. Of the 1,602 DES objects in this interval, 152 matches to the Pantheon+ inventory and two ambiguous associations are excluded. This leaves 1,448 objects: a set distinct from the calibration set. Distinct objects do not guarantee independent standardization and calibration. An unknown cross-covariance between the samples is assumed zero, not proved absent.

The local protocol was frozen on 13 September 2026 at 04:56:05 UTC, before acquiring the individual data and calculating the new likelihood. The DES publication's aggregate results were already known. There was no external preregistration or blinding to the available literature.

Following the DES convention, the distance-modulus shape to compare is:

$$\tilde{\mu} = 5\log_{10}\left[(1+z_{HEL})\int_0^{z_{HD}} \frac{d\zeta}{\mathcal{E}(\zeta)}\right]. \tag{G1}$$

The two redshifts have different reference-frame corrections; the additive constant absorbs absolute calibration. For residuals $r = y - \tilde{\mu}$ and covariance C, the only refitted parameter in the primary transfer is:

$$\hat{b} = \frac{\mathbf{1}^T C^{-1} r}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \chi^2 = (r - \hat{b}\mathbf{1})^T C^{-1} (r - \hat{b}\mathbf{1}). \tag{G2}$$

The NPZ files contain precision, the inverse of covariance: the full matrix is inverted before selecting the covariance submatrix. One neither selects the precision submatrix directly nor adds the error diagonal again. Source and commit: [10].

H | Statistical diagnostics and the solution's limit

Why we give no validation percentage

The compressed DES likelihood incorporates weights related in part to supernova classification. In the local study's Gaussian surrogate, which propagates uncertainty in the calibrated matter parameter, the ratio $1265.0414/1447$ is about 0.87425; the lower-tail probability is about 0.0002094. Both controls have values of the same order. These are surrogate diagnostics, not calibrated p-values for the full DES procedure or probabilities that MM is false or true.

The sum of Ia probabilities in the study's pinned full-release files is 1714.22219; in the selected sample it is 1343.86677. The effective number 1684 reported in the DES paper is not reproduced by those sums. The exact convention or provenance of the difference remains unresolved; it is not replaced by a convenient choice of degrees of freedom.

A separate check on the full sample with the standard model, including the official offset-marginalization term, reproduces the published value 1640.3 to its tabulated precision. This supports the likelihood implementation; it does not provide an absolute calibration of B17's significance. None of these checks extends B17 beyond its domain.

Where backward continuation stops

For $p = 1/4$, $a_i = 1/2$ and $m_i = (1/2, 1/2)^T$, set $u = 3\log(a/a_i)$. The continuous solution for component B can be written:

$$m_B(u) = \lambda_+ e^{\lambda_+ u} + \lambda_- e^{\lambda_- u}, \quad \lambda_\pm = \frac{1 \pm \sqrt{5}}{4}. \tag{H1}$$

Its first zero when evolving backwards occurs at:

$$u_* = \frac{\log(-\lambda_-/\lambda_+)}{\lambda_+ - \lambda_-}, \quad a_* = \frac{1}{2} e^{u_*/3}, \quad z_* = a_*^{-1} - 1. \tag{H2}$$

Numerically, $z_* \simeq 1.66468666$. Continuing the same formula further into the past makes m_B negative. This is not a physical state allowed by the count model. The comparison remains restricted to $z \leq 1$, where the solution is admissible.

The D1 values in the frozen curve are $w_D(1) \simeq -0.81514356$ and $w_D(a_i) = -0.875$. The dominant-mode-only example instead has $w = -\lambda_+ \simeq -0.80901699$. These values are not interchangeable. Changing the initial state or the past extension would require declaring a new version and repeating the relevant comparisons.

I | A candidate memory-space rule

This appendix defines an exploratory model distinct from the three-type B17 process in B-F. Use S for spatial registers, M_{mem} for memory records and $N = S + M_{\mathrm{mem}}$. A spatial register carries a fixed symbol x_i from an alphabet of m symbols. A memory is retained as a record but is excluded from the spatial adjacency graph by the candidate rule. Copying memories is a specified operation, not a derivation of an observer or of quantum measurement.

Write r_e for the reading channel on an edge. It is the quantity called q in the exploratory notes [16], not the epoch channel q_n of Equation 1. Independently for the two endpoints, a reading copies the stored symbol with probability r_e and otherwise draws a uniform symbol. Therefore:

$$A_r = \frac{1 - r^2}{m} + r^2, \quad B_r = \frac{1 - r^2}{m}, \quad \Pr(\text{confirm} \mid x_i, x_j) = B_r + r^2 \mathbf{1}_{x_i = x_j}. \tag{I1}$$

[CANDIDATE MM RULE] Confirmation creates one memory. Different readings request a spatial interposition. Each prerequisite comparison is processed by the same reading rule. A request that cannot be completed creates one memory of an unrealized distinction. A realized request creates one spatial register with a fresh uniform symbol. Earlier realized prerequisites remain, even if a later request fails. Thus every reading has its own classified record; the entire chain is not a single indivisible update.

The numerical implementation uses ordered tetrahedral bisections and compatibility between adjacent cells. Three-dimensional cells, their ordering, the selection weights and the reading channel are supplied rules. Spatial distance means shortest path length in the current spatial graph. It is not the negative logarithm of a historical correlation, and neither construction is identified here with a relativistic metric.

In a region initialized with equal symbols and $r_e = 1$ on internal links, every internal reading confirms. This protects those links from subdivision. Persistence of the reported internal distances is a numerical result for the stated mesh preparation; a general theorem excluding all changes of shortest paths through the exterior is not claimed.

The limit $r_e = 0$ gives confirmation probability $1/m$ regardless of the stored states or previous readings. It recovers B17's independent coincidence probability, not its three-type transition kernel, energy assignment or cosmological predictions.

J | Event accounting and mean growth

Let K count top-level attempts. At a given state, choose a cell event with probability $s = S/N$; otherwise copy a memory, adding one memory. Conditional on a cell event and the complete current state, define γ_S and γ_{mem} as the expected spatial and memory increments, including all prerequisite readings. They can depend on the state and need not be constants.

If R_{done} counts realized readings, F confirmations, D_{stop} unrealized distinctions and L memory-copy events, the bookkeeping identities are:

$$S - S_0 = R_{\mathrm{done}}, \qquad M_{\mathrm{mem}} - M_{\mathrm{mem},0} = F + D_{\mathrm{stop}} + L. \tag{J1}$$

The numerical v2 model satisfies these identities in the audited runs. One cell event can produce several records; it is incorrect to assume that every top-level attempt increments N by exactly one. The conditional drifts are:

$$\mathbb{E}[\Delta S] = s\gamma_S, \qquad \mathbb{E}[\Delta N] = 1 + s(\gamma_S + \gamma_{\mathrm{mem}} - 1). \tag{J2}$$

Expectations in J2 are conditional on that state. They follow directly by separating cell events from memory-copy events. A deterministic mean-drift approximation then gives:

$$\frac{ds}{dN} = \frac{1}{N}\left[\frac{s\gamma_S}{1 + s(\gamma_S + \gamma_{\mathrm{mem}} - 1)} - s\right]. \tag{J3}$$

J3 is a coarse-grained approximation, not an exact equation for the mean of a stochastic ratio. An adequate fluid limit, moment control and suitable regularity are required before treating it as a convergence theorem. With stationary coefficients and a memory-dominated regime $s \to 0$, it gives $d\log S/d\log N \to \gamma_S$. This statement is conditional; transient measured coefficients do not by themselves establish an asymptotic exponent.

In the reference control at $N = 131072$, $S_0 = 64$ and $S = 27132$. The 27068 additions split into 17241 initial-request and 9827 prerequisite realizations. Its 103940 memories split into 11517 confirmations, 4421 unrealized distinctions and 88002 copies [16]. These are checks of the classified process, not observational counts.

Using coefficients measured within the same evolving run reconstructs its growth only approximately. The recorded maximum discrepancy in S/N is 0.03042. The next appendix supplies a distinct calculation of the coefficients from an idealized chain law.

K | Closed equations for independent chains

Let J be the number of prerequisite requests needed to complete a distinguishing node if all prerequisites succeed. Use distributions π^{top} for initial requests and π^{f} for forced requests. This J is called K in the chain notes; it is not an energy count. Assume independent child subtrees, independence of J from reading outcomes, and a constant confirmation probability p, with $\alpha = 1 - p$.

Define the generating function and the expected attempted-prefix factor:

$$G_f(R) = \sum_{j \geq 0} \pi_j^f R^j, \qquad H_f(R) = \sum_{j \geq 0} \pi_j^f \sum_{h=0}^{j-1} R^h. \tag{K1}$$

The inner sum is zero when j is zero. Define G_{top} and H_{top} in the same way. If R is the success probability of a forced request, its own reading must distinguish and all its children must succeed. Hence $R = \alpha G_f(R)$. Use the probabilistically relevant fixed point obtained by finite-depth recursion; in the subcritical case used here it is unique.

Let T_f be the expected number of readings in a forced subtree and M_f its expected number of spatial realizations. Processing stops at the first failed child, but completed earlier children remain. Consequently:

$$R = \alpha G_f(R), \qquad T_f = \frac{1}{1 - \alpha H_f(R)}, \qquad M_f = R T_f. \tag{K2}$$

For an initial cell event the spatial and memory yields are:

$$\gamma_S = \alpha[G_{\text{top}}(R) + H_{\text{top}}(R) M_f], \quad \gamma_{\text{mem}} = 1 + \alpha H_{\text{top}}(R) T_f - \gamma_S. \tag{K3}$$

These are exact expectations for the independently generated chain model. Finiteness requires the relevant expected subtree size to be finite. A sufficient condition is $\alpha \mathbb{E}[J_f] < 1$, satisfied by the input distribution used here. The formulas do not assert independence in the spatial mesh; applying them to that mesh is an approximation to be assessed.

No metric, gravitational constant or cosmic clock enters K1-K3. Their contribution is a conditional relation between local prerequisite statistics and mean production of spatial and memory records.

L | Proof and historical short-run comparison

For fixed J, child number h is reached only if the preceding h-1 children succeed. Its reach probability is R^{h-1}. Summing and averaging over J gives H. Independence lets each reached child contribute its unconditional expected reading count and spatial yield. Thus $T_f = 1 + \alpha H_f T_f$ and $M_f = R + \alpha H_f M_f$, proving K2. Applying the same argument to the initial request gives K3; every reading becomes either a spatial record or a memory.

The historical geometric-only sample contains 4965 initial requests and 3163 forced requests, ending at 8192 spatial registers. Its mean required child counts are 0.533736 and 0.162188. These measured distributions are provided in the English evidence file [16]. With $p = 1/4$, the equations give $R = 0.717896$, $\gamma_S = 0.939370$ and $\gamma_{\mathrm{mem}} = 0.465357$.

Check	Spatial yield / memory yield
Closed independent-chain equations	0.939370 / 0.465357
Separate 150000-chain simulation	0.940153 / 0.464367
Monte Carlo standard errors	0.002351 / 0.001839
Recorded spatial-mesh control	0.941234 / 0.554211

The separate simulation checks the idealized equations, not the complete mesh. The approximately 16 percent memory deficit belongs to this short-run comparison; it is not an asymptotic error estimate. The spatial agreement does not establish universal 0.2 percent accuracy. Appendix Q now tests a specified reading process on completed levels; its cohorts and inputs differ, so its numbers do not silently replace this table.

Required length must be distinguished from visited length. If two children are always required and each succeeds with probability one half, the expected number visited is 1.5. Substituting the already stopped visited-length distribution back into H gives 1.25, counting the stopping effect twice. The existing measurement under noisy readings stops on failure, so its distribution cannot simply replace the required-length law. Dependence on reading history is another possible contribution, not an isolated explanation of the entire deficit.

The source-hashed population approximation gives $S/N = 0.20719$ against 0.20700 in the recorded control. Its recorded final cell weight is 384, equal to its initial weight; spatial and memory yields are 0.94037 and 0.47909. Revision 03 mistakenly described an earlier, nonconserving version rather than this corrected source. The approximation remains auxiliary: close agreement in one fraction does not establish general predictive accuracy. Neither the proof above nor the completed-level tests in Q depend on it.

M | Reading history and spatial localization

With A_r, B_r from I1, the exact next confirmation probability is $B_r + (A_r - B_r)\Pr(x_i = x_j \mid \mathcal{F})$, where $\mathcal{F}$ includes the relevant available network history and selection information. A posterior computed from only one edge's confirmations and distinctions need not equal this full-network posterior.

For an initially independent four-symbol pair with $r = 1/2$, one confirmation changes the next confirmation probability from 0.25 to 0.296875. In a triangle, confirmations on AB and BC change the first confirmation probability on AC to 0.26171875. Direct enumeration proves both examples. At $r = 0$, I1 instead makes the probability exactly $1/m$ even after conditioning on the previous network history.

In the recorded coherent-region sample, enumeration on common neighbours changes the mean prediction of 419 rereadings by only 0.00109. The mean absolute change is 0.00662 and the largest is 0.11412. Positive and negative changes cancel in the mean. This is not a 0.001 error bound. The calculation omits longer network paths and 45 of the 1500 sampled readings because the selected neighbourhood was too large. It is a restricted diagnostic, not a complete posterior calculation.

For localization, write D_{stop} for unrealized distinctions; this is the U counter of the source notes, not physical energy U in Appendix D. Assign each such record to the smaller current graph distance of its two source endpoints from the initial region. Divide by the spatial-register count in that shell. At $N = 65536$, the shell-zero and shell-one ratios are 106.875 and 9.786 in the coherent run, versus 4.25 and 0.245 in the control [16].

These data support strong near-region accumulation under that localization convention. They do not establish exact confinement to two shells or equality of the outer profiles: at distance three the ratios are 0.240 and 0.125. Shared graph distances are not automatically shared physical radii, and a single paired run does not remove all effects of age, geometry or initialization.

The quantities are counts in an exploratory network. A mass profile, pressure field, causal transport law or lensing signal would require additional physics and independent comparisons. No dark-matter interpretation or exclusive MM observational signature is inferred from this accumulation.

N | Energy assignments without changing B17

Introduce K_E as an effective energy count for the exploratory spatial model. It is not automatically the active count of B17. With positive calibration constants v and epsilon, adopt:

$$V = vS, \qquad E_D = \varepsilon K_E, \qquad P_D = -\frac{dE_D}{dV}, \qquad w_D = -\frac{d\log K_E}{d\log S}. \tag{N1}$$

[IMPORTED EFFECTIVE POSTULATE] N1 assigns physical volume and energy to relational counts and assumes a homogeneous component with no additional energy exchange. Its last identity follows by dividing the work balance by $\rho_D = E_D/V$. Applying it to smoothed counts requires a specified coarse graining; it is not an instantaneous pressure assigned to an individual random jump.

The minimal assignment $K_E = S$ yields constant density ε/v and $w_D = -1$. This is a conditional microscopic interpretation of a constant-density component. It neither fixes that density's observed value nor explains the separate vacuum residue in the inherited cosmology.

For the alternative $K_E = S + \eta D_{\mathrm{stop}}$, with constant nonnegative relative energy weight eta:

$$w_D = -1 - \frac{d\log(1 + \eta D_{\mathrm{stop}}/S)}{d\log S}. \tag{N2}$$

With $\eta = 1$, logarithmic secants over the recorded interval from 32768 to 131072 total records give approximately -1.0242 in the control and -1.0844 in the coherent run. These are finite-interval slopes of candidate energy counts. They are not measured cosmological equations of state, and a localized contribution need not admit the homogeneous description in N1 without further work.

Assigning energy to all M_{mem} records is a different alternative: D_{stop} is only a subset. Coupling a memory sector to other matter would also add exchange terms and invalidate a direct inference of its pressure from the no-exchange balance alone.

General relativity specifies how an assigned energy-momentum tensor sources geometry; it does not set the energy of a record to zero because that record adds no spatial volume [15]. Thus $K_E = S$ is a minimal effective MM choice, not a consequence of importing general relativity. Both the assignment and its coupling must remain explicit.

The original volume-capacity and active-energy dictionary in D-F, the frozen parameters in E and G, and every result in Table 1 remain unchanged. No numerical result of that observational comparison validates N1-N2 or selects between these new microscopic assignments. This extension contributes a clearer candidate mechanism and a conditional growth calculation, not a silently revised cosmology.

O | An explicit state-dependent event law

Let η contain the ordered mesh, each spatial register's retained symbol, the channels and the memory count. The internal recursion also has an ordered request stack. A reading uses I1 conditional on the actual endpoint symbols. It need not reconstruct a posterior from a shortened record of earlier readings. Marginalizing this process over unobserved symbols would require the corresponding full conditional distribution.

[CANDIDATE MM RULE] In the homogeneous implementation, choose a tetrahedron c of generation g with weight proportional to 2^{-g}. Resolve the oldest incompatible prerequisite first, breaking equal-generation ties by increasing cell identifier. Use the ordered two-child bisection specified in the supplied program. Each realized interposition draws a fresh uniform symbol. Confirmations and failed distinctions produce the records defined in I. These choices specify the candidate process; they are not uniquely forced by ontology.

For a terminating cell event, let h run over its possible reading and new-symbol histories. Its probability $P_\eta(h \mid c)$ is the product of the conditional reading factors and fresh-symbol factors along that history. Write $R_S(h)$ and $R_M(h)$ for its two record outputs. Then:

$$\Gamma_X(\eta) = \sum_c \omega_c(\eta) \sum_h P_\eta(h \mid c) R_X(h), \qquad X \in \{S, M\}. \tag{O1}$$

Here the cell weights sum to one. For almost-surely terminating events with integrable outputs, O1 is the exact conditional expectation: multiply each possible outcome by its probability and sum. In J2, the two growth coefficients are precisely these state-dependent quantities. The formula defines an explicit expectation, not a cheap evaluation algorithm or a two-variable closure.

If $\mathcal{P}$ includes cell selection and memory copying, the distribution of full states obeys:

$$\mu_{k+1}(\eta') = \sum_\eta \mu_k(\eta) \mathcal{P}(\eta, \eta'). \tag{O2}$$

The elementary reading-and-stack process is specified even before event outputs are grouped. O1-O2 at the grouped-event level require the stated termination conditions. The finite runs verify termination for every executed event, not a theorem for every infinite refinement history. Memory copies leave mesh and symbols unchanged, so removing them gives exactly the embedded chain tested in Q; it does not preserve total-record growth exponents or physical time.

P | Why counts alone do not close the model

There is a simple obstruction to replacing the state-dependent law with an exact function of S and memory count only. Start with the same periodic three-by-three-by-three mesh, the same channels $r = 1/2$ and the same counts. All initial reference edges are compatible, so their events need no prerequisites.

In preparation A give every vertex the same symbol. In preparation B use the first lattice index modulo three as the symbol. Each initial reference edge joins opposite cube corners, so its symbols differ in B, including across periodic boundaries. Both preparations are allowed by the four-symbol state space. Every selected initial event has the same respective confirmation probability within each preparation. Equation I1 therefore gives:

$$\Gamma_S(\eta_A) = \frac{9}{16}, \qquad \Gamma_S(\eta_B) = \frac{13}{16}, \qquad (S, M)_A = (S, M)_B. \tag{P1}$$

Thus an exact autonomous drift depending only on these two counts cannot describe all allowed states. This is a counterexample, not a numerical lack of precision. It does not exclude an approximate closure for a restricted ensemble, or additional variables that summarize the relevant state correlations. No universal reduced closure is asserted for the particular randomly initialized ensemble either.

Event accounting can also be checked without a simulation. Take an initial request with two ordered leaf prerequisites and let their distinction probabilities be alpha-zero, alpha-one and alpha-two. There are four possible stopped histories: immediate confirmation; first-child confirmation; first-child success followed by second-child confirmation; and complete success. Their spatial/memory outputs are respectively 0/1, 0/2, 1/2 and 3/0.

Weighting these outputs gives mean spatial yield alpha-zero times alpha-one, multiplied by (one plus twice alpha-two). Mean total readings are one plus alpha-zero multiplied by (one plus alpha-one). The supplied check enumerates all four paths for 125 probability triples, including zero and one. Earlier successes are retained in the third history; a global rollback would implement a different model.

The gain is structural: the reading rule and the full state determine growth, while compressed descriptions have an explicit job and a measurable approximation error. This is a mathematical completion of the specified finite-event formulation, not a derivation of the rule, dimension, energy unit or cosmological clock.

Q | Completed-level reading-network tests

A separate compact implementation tests the homogeneous candidate process with four symbols and channels $r = 0$ or $r = 1/2$. For each channel it uses four independent random seeds, 301-304, on an initial periodic mesh with 27 vertices and 162 tetrahedra. Each run continues until refinement triplet k=2 has registered all 1728 D, 5184 F and 5184 E rings: 12096 distinct realized interpositions. All eight runs reach this criterion.

Reported yields use the cohort of initial requests whose generation belongs to triplet 2, including every prerequisite reading they trigger, even from an older triplet. They are per cell event, not per spatial registration or per cosmic-time interval. Each run has equal weight in the means; brackets give the range across its four seeds, not a confidence interval.

Channel and measured quantity	Mean [seed range]
Memoryless: spatial yield	0.92828 [0.92360, 0.93543]
Memoryless: memory yield	0.43216 [0.42798, 0.43514]
Memoryless: confirmation share	0.24947 [0.24848, 0.25163]
Half-signal: spatial yield	0.89297 [0.88915, 0.89569]
Half-signal: memory yield	0.45808 [0.44954, 0.46554]
Half-signal: confirmation share	0.26593 [0.26316, 0.26922]

Every cell event satisfies the record balance. Total cell-selection weight is invariant. A geometry-only closure probe measures required children and then restores the original mesh and identifiers without drawing randomness. The tests check restoration, reading-channel marginalization, completed counts and result fingerprints. They do not establish equivalence to every earlier prototype; the tie-breaking rule is specified explicitly here.

Using each run's required-child distributions and measured confirmation share, K predicts mean spatial/memory yields 0.92783/0.43078 and 0.89387/0.45207. This is a same-run diagnostic, not autonomous prediction. Keeping one quarter instead in the half-signal case gives 0.92600/0.43025: the old probability is not transferable unchanged.

Larger initial-mesh trials are also retained. Six of eight stop at a declared 80000-spatial-register resource cap before completing triplet 2; none is used as a completed cohort here. Four seeds at one completed size are not an infinite-size or asymptotic test. Coherent regions, their formation, memory copying and energy assignments are outside these new runs. No observed-data fit is added by them.

R | Pair audit and present boundary

The later pair-and-cluster approximation in the source work improves several geometric predictions [16]. For example, its reported forced fractions for D/F/E are approximately 0.370/0.610/0.502 against 0.370/0.611/0.500 in the geometric reference. This is useful evidence for retaining correlations. It is not a validation of the full reading network in Q.

A further audit checks whether each four-entry pair table has the same one-node marginals as the separately evolved common marginals. A consistent family representing those same variables must satisfy both requirements. Normalization alone is insufficient.

The audited tables stay nonnegative and normalize to within about one part in ten trillion. Nevertheless, during active evolution the largest pair/common-marginal discrepancy is 0.010728 with 1500 integration steps and 0.010749 with 3000. The tested interval requires both common marginals to lie between 0.05 and 0.95. The discrepancy persists under this refinement; it is not explained by normalization roundoff. No general convergence claim follows from only two resolutions.

Consequently these tables remain an approximate closure and cannot be treated as one exact joint probability model with the quoted common marginals. Conditioning on several known neighbouring registrations also requires their joint probability; dividing separately by each marginal is an additional factorization assumption, not exact Bayes conditioning without qualification. This limitation does not invalidate the explicit reading law in O or its tests in Q.

The reported pair-based memory yields 0.4531 and 0.4553 differ by about 0.48 percent, not less than 0.25 percent. Both are outputs of a chain transfer comparison, not a demonstrated accuracy for noisy-network memory production. Cohorts, reading histories and geometric averages must be matched before making such comparisons.

The archive now contains the English state-dependent implementation, eight complete run records, the capped larger trials, and 39 portable algebra/accounting/evidence checks. These 39 checks recheck the supplied results; the separate simulation command regenerates the trajectories. The pair audit records are accompanied by source hashes and an English audit specification; the original Italian pair prototype is retained locally, not included as a portable English implementation.

The specified finite-event mechanism, its conditional expectation and the bounded homogeneous test are complete. A universally accurate reduced closure, physical calibration, formation of coherent matter and unique derivation from MM remain open. B17 and its observational comparison are unchanged; this work is not exclusive observational evidence for MM.

S | The model against the three geometric probes at once

This appendix and the three that follow confront the relational construction with the three standard geometric probes of cosmology at the same time, with the standard cosmology free rather than frozen. The outcome, stated before the details, is threefold. The model passes: at the minimum of the profile likelihood the 1,590 supernova light curves, the thirteen baryon-oscillation observables and the three compressed microwave numbers are fitted together with a total chi-square of 1420.05, as well as a cosmological constant does, and no block is sacrificed to any other; the sampled chain of Appendix T reaches 1420.02, a numerical rather than physical difference. Standard cosmological parameter values emerge from the fit rather than being imposed on it, and the profile best fit lies at $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$, each within about one standard deviation of the Planck 2018 values, inside the adopted Friedmann-Lemaitre-Robertson-Walker and recombination framework and with no quantity imposed beyond the photon temperature, the neutrino content and the branching law itself. And its one free amplitude, a convention until now, is bounded rather than assumed: p, the departure of the relational component from a cosmological constant, is held below a few hundredths, which excludes the larger value the restricted comparison of Appendices G and H holds fixed.

The construction also commits itself. By the law of (S1) the relational dark component satisfies $\bar{w}_D \geq -1$ at every epoch, with any vacuum fraction: it cannot cross into the phantom regime. The present data need no such crossing, and the fit is comfortable without one. A future release that establishes a crossing at a significance it declares decisive would exclude this model in this form. That is a commitment made in advance, not a comment made afterwards.

Appendices G and H compare a frozen curve with supernovae over a stated late-time domain. That comparison cannot reach the cosmic microwave background, and the reason is structural rather than practical. The solution used there starts from an initial state at $a_i = 1/2$ and, continued backwards, reaches a first zero of the growing component at $z_* \simeq 1.66$, beyond which the count model admits no state. A construction that ends at redshift about one and a half has no recombination epoch to compare with.

The comparison in this appendix uses a different object: the same branching law, switched on locally and late by the gate of Appendix O, integrated as a background from the network seed to the present. Before switch-on the dark component is inert and the expansion is the standard one, so the model has an ordinary early universe, an ordinary recombination, and therefore a sound horizon and an acoustic scale that can be confronted with the compressed measurements. The gate is what makes the question askable; it is not a device introduced to fit the data.

Three observational blocks enter, none of them refitted or reweighted here. The supernovae are the Pantheon+ compilation above redshift 0.01 [19], without the SH0ES calibration: 1,590 light curves of 1,473 distinct supernovae, duplicates kept with the published statistical and systematic covariance, and a single brightness offset removed analytically. This is not the DES sample of Appendix G. Its 1,448 objects coincide in number with the redshift-one Pantheon+ selection of B18 and with nothing used here: the global comparison takes the full Pantheon+ redshift range. The baryon acoustic oscillations are the DESI DR2 compressed measurements [9]. The microwave background enters only through the Planck 2018 distance priors, the shift parameter R, the acoustic scale ℓ_A and the baryon density, with their correlations [17]. These three numbers summarize the geometry of the last-scattering surface; they are not the temperature and polarization likelihood. The distinction matters here more than usual and is stated again in the limits below.

The background is solved in closed form. The count equation is linear with a scalar coefficient, so its solution is the exponential of the integrated gate acting on the seed state, and the dark fraction, the expansion rate, the comoving distances and the sound horizons follow by quadrature from densities fixed to the standard ones. Recombination is the original HyRec integration on the actual expansion history. The only emulated quantities are the two recombination redshifts and the certificate blocks; their declared errors enter the covariances rather than being neglected. The certification refits the emulator 128 times, each time without the point it then has to predict. No false acceptance occurs in any of its 4,608 decisions, 4,572 of them inside the certified domain; every supernova row stays within one millionth of a standard deviation, every baryon-oscillation row within 0.24 per cent of one and every microwave row within 0.005 per cent; and inside the domain 4.1 per cent of the cases that should be accepted are rejected, which is the conservative direction. The certificate holds only inside that declared domain.

$$\bar{w}_D = -1 + s(u)\, p\, f(u), \qquad f = \frac{m_B}{m_A + m_B}, \tag{S1}$$

with s the gate, zero before switch-on and one after it. Equation (S1) already shows what the data can constrain: not p alone, but the amplitude of the whole product. A small p, a gate that opens late, or a dark sector only partly made of the branching component all reduce the same quantity.

T | What the comparison bounds

In brief	Value
Bound on p, one-sided, 95 per cent	0.0724, from the converged sampled chain
Lowest total chi-square of the sampled chain	1420.02: supernovae 1403.87, baryon oscillations 10.98, microwave background 5.17
Minimum of the profile likelihood	1420.05: supernovae 1403.4, baryon oscillations 11.6, microwave background 5.1
Profile best fit	$h = 0.678$, $\Omega_m = 0.306$
Posterior medians of the chain	$h = 0.6764$, $\Omega_m = 0.3066$
Cosmological constant, same code and same data	1421.19
Main methodological limit	the microwave background enters as three distance priors, not as the full temperature and polarization likelihood

The sampled parameters are the three transition parameters $(p, u_c, \Delta u)$, the standard cosmology (h, ω_b, Ω_m) and the two spatial parameters of Appendix U. What the dark sector is free to do is contained in those three numbers: the branching probability, the epoch at which the gate opens and its width. There is no free function of redshift here, no expansion of $w(z)$ in powers or nodes, and no reconstruction: once the three are fixed, (S1) and the count equation determine the whole history, including recombination. A comparison of this kind can fail in a way a flexible parameterization cannot. The result is stated first as a profile, because a profile is easier to check than a posterior: for each value of p the total chi-square is minimized over the standard cosmology, with the gate free.

p	best total χ^2	difference from the minimum
0.25	1468.49	+48.44
0.20	1451.94	+31.88
0.15	1437.64	+17.59
0.10	1426.77	+6.72
0.05	1420.75	+0.70

p	**best total χ^2**	**difference from the minimum**
0.03	1420.05	0
0.01	1420.51	+0.46

Each row is minimized over the gate as well: a late switch-on is mildly preferred, worth about six units of chi-square at $p = 1/4$ against the two earlier epochs tested, and the gate width is worth about one. The ordering in p is the same at every epoch, because the background observables respond to the product in (S1) rather than to its timing alone. The minimum is an acceptable fit, with the supernova, baryon-oscillation and microwave contributions at 1403.4, 11.6 and 5.1, reached at $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$; the one-parameter interval of $\Delta\chi^2 = 3.84$ closes near $p \simeq 0.08$.

The reference to compare it with is obtained from the same code. Setting the vacuum fraction of equation (8) to one removes the branching component entirely and leaves a cosmological constant, evaluated with the same closed-form background, the same recombination, the same distance priors in the same convention and the same three data blocks, with the cosmology equally free; that gives 1421.19, at $h = 0.6837$, $\omega_b = 0.02253$ and $\Omega_m = 0.3014$, on three free parameters against the model's six. The model's profile minimum, 1420.05, sits about one unit of chi-square below that, on 1,606 measurements. The dark sector here is free in three numbers, however, against none for a cosmological constant, so one unit is not a gain per parameter: any information criterion prefers the simpler model, and this comparison provides no evidence in favour of the relational construction over a constant. What it does establish is the other half of the statement, and it is not a small one: a dark component derived from a counting law, with no freedom to choose the shape of its history, meets the reference on the reference's own ground, across supernovae, baryon acoustic oscillations and the geometry of last scattering at the same time.

The value $p = 1/4$ used in the frozen curve of Appendices G and H is therefore disfavoured by $\Delta\chi^2 \simeq 48$ once the three blocks are combined. The penalty is shared rather than carried by the block added last: between the minimum and $p = 1/4$ the supernovae move from 1403.4 to 1418.8, the baryon oscillations from 11.6 to 25.1 and the compressed microwave background from 5.1 to 24.6, while the preferred expansion rate falls from $h = 0.678$ to 0.643. This is a statement about that frozen configuration, not a refutation of the branching law: the model contains a second way to reduce the amplitude in (S1).

That second way is the vacuum fraction ν_v of equation (8), the share of the dark sector that is a genuine constant rather than the branching component. It is fixed to zero throughout the rest of this work, which is the strong reading: all of dark energy is relational. Releasing it, with the gate held at the middle epoch of the table above, gives

ν_V	$\Delta\chi^2$ at $p = 1/4$	$\Delta\chi^2$ at $p = 0.03$
0	+54.04	0
0.50	+13.52	+0.07
0.80	+1.75	+0.44
0.90	+0.52	+0.62
0.95	+0.47	+0.72

The differences in this second table are taken from its own minimum, reached at $\nu_V = 0$ and $p = 0.03$ with that fixed gate; they are not directly comparable with the first table, whose rows are minimized over the gate as well. At $\nu_V = 0.90$ the combination $(1 - \nu_V)p = 0.025$ reproduces the chi-square of $p = 0.03$ at $\nu_V = 0$, and above that fraction the two columns coincide: the model has become a cosmological constant and p is no longer constrained at all. The honest formulation of the result is therefore

$$(1 - \nu_V)\, p \lesssim 0.03 \text{ to } 0.07, \tag{T1}$$

a bound on how far the relational component may depart from a cosmological constant, weighted by its share of dark energy; it is not a bound on that share. Under the strong reading $\nu_V = 0$ all of dark energy is relational and (T1) is a bound on p; under a weak reading it permits $p = 1/4$ at the price of attributing nine tenths of dark energy to an ordinary constant, which is a different theory from the one this article proposes.

The sampled chain answers the marginal question, and it converged on its own terms. With 32 walkers and a differential-evolution ensemble move it ran to 8,000 steps, at which point it exceeded the declared rule by a wide margin: the largest autocorrelation time is 103 steps, so the chain is 77 times longer than its own memory against the 50 required, and every autocorrelation time had settled to within two per cent between checks. After discarding five autocorrelation times and thinning, 8,896 samples remain, about 2,300 of them independent. The marginal posterior of p under the strong reading gives one-sided upper limits of 0.043 at 68 per cent, 0.072 at 95 per cent and 0.097 at 99 per cent, with a median of 0.033 against a lower prior edge of 0.01. The standard cosmology comes out at $h = 0.6764$, $\omega_b = 0.02258$ and $\Omega_m = 0.3066$, with 68 per cent intervals $[0.6714, 0.6806]$, $[0.02246, 0.02271]$ and $[0.3022, 0.3114]$. The gate epoch and its width span their priors, and so do the two spatial parameters: the comparison does not constrain when the transition switches on, only how large it may be. The lowest total chi-square the chain met, 1420.017 with 1403.87 from the supernovae, 10.98 from the baryon oscillations and 5.17 from the microwave background, agrees with the minimum of the profile in the first table, 1420.05, although the two were computed by different routes: the profile from the closed-form background directly, the chain through the certified emulator. The chain's best point lies between the nodes of the profile grid, at a slightly later and sharper

switch-on, which accounts for the difference of 0.03: it is numerical, not physical. The marginal limit and the profile interval answer different questions and are reported separately. Two further chains are consistent with the bound without adding to its weight. One, on the same certified emulator with the default stretch move, was stopped at 11,500 steps, 42.7 autocorrelation times against the 50 the rule requires, and gives between 0.073 and 0.076 at 95 per cent depending on burn-in and thinning. The other, an earlier pilot on twelve of the 128 certificate points run for 3,000 steps as a test of the machinery, gives 0.072. Both are recorded in the research materials.

Why the bound is physically informative. A bound fixes where the model may lie; the agreements below are what keep it there, and they are reported with the same weight. First, the three blocks agree with one another at the minimum: the supernovae give 1403.4 for 1,589 degrees of freedom, the baryon oscillations 11.6 for 13, the compressed microwave background 5.1 for 3, and no block pulls against the others once the standard cosmology is free. Second, the standard cosmology at the profile best fit, $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$, lies within about one standard deviation of the Planck 2018 values from three blocks that share no systematics, with the transition present. Third, the degeneracy that equation (S1) predicts is the one the data exhibit: the vacuum fraction and p trade against each other along $(1 - \nu_v)p$ to within the second decimal, which is a structural prediction of the model confirmed rather than a fitted feature. Fourth, the branching law forbids a phantom crossing, and the minimum needs none. Fifth, where the data express a preference on the timing they lean the way the local-transition picture leans, towards a late switch-on, although only at the level of a few units of chi-square at large p and less than one at the minimum, which is noted and not built upon. None of these is a detection. Separately each is modest; together they are a concordance that a bound alone does not provide, and they are reported here with the same weight as the bound.

Scope of the comparison. The microwave background enters as three compressed numbers in the convention of their own derivation, which reproduces the published shift parameter to 0.03 and the acoustic scale to 0.6 of their standard deviations but differs by about three from a direct angular scale taken from a Boltzmann code; the convention is part of the measurement and is used as published, not mixed with another. The late integrated Sachs-Wolfe effect of a transition switching on at low redshift is not covered by a geometric compression, and no temperature or polarization spectrum is computed. The domain of the comparison is the box declared by the emulator certificate, $p \le 0.3$, and nothing is claimed outside it.

U | The negligible-noise branch, and the threshold below it

The expansion history above is a mean. The gate is local, so regions switch on at different times, and a finite number of records per coherence cell makes the mean itself fluctuate. The analysis handles this by bounding rather than modelling: for each sample the certificate computes an upper bound on the chi-square that the omitted count noise could contribute, and the sample is used only if that bound stays below a declared budget. The variance at switch-on is exact,

$$\mathrm{Var}(\delta \ln N) = \frac{2 - (a_{\mathrm{seed}}/a_{\mathrm{on}})^3}{N_{\mathrm{seed}}}, \tag{U1}$$

so the bound falls as one over the number of records per cell at the seed. At a reference point, $N_{\mathrm{seed}} = 10^{20}$ with a coherence length of 100 Mpc, the bound is 4×10^{-11} in chi-square, eleven orders of magnitude below anything the three blocks can resolve. That point is the corner of the chain's starting region, not an edge of the prior, which extends to $N_{\mathrm{seed}} = 10^4$: 53 per cent of the retained samples lie below 10^{20}, the smallest at about 1.7×10^9, each kept in the negligible-noise branch by the certificate below, not by the prior. The two spatial parameters are unconstrained by the comparison; this is not the data preferring a fine-grained network.

Reading (U1) in the other direction gives the only quantitative target this work can offer for the inhomogeneous sector. The omitted contribution reaches a hundredth, a tenth and one unit of chi-square at about 4×10^{11}, 4×10^{10} and 4×10^9 records per cell respectively, and the declared budget is crossed near 10^{11}. These values hold at the reference coherence length of 100 Mpc and move with the other parameters, which is why the certificate works sample by sample; the interesting region, down to 10^4 records, is wide and mostly below the branch used here.

What happens there is not covered by this analysis, and the code refuses those samples rather than extrapolating: a non-negligible noise requires the full cross-dataset covariance with its determinant, not an omission bound. The chain applies the same refusal when a sample's count noise would imprint on the microwave temperature or its lensing beyond the declared limits, because three distance priors cannot absorb such an imprint. Of the production chain's 256,064 evaluations, 12,433 were refused for that reason and 791 for an omission bound above the budget; 168,273 were proposals outside the prior box, which a differential-evolution ensemble makes routinely, 333 fell outside the emulator's design and 4 failed the late gate-closure check. The counts are in the deposited chain summary. Any lower edge a sampled chain shows in the number of records per cell is consequently a boundary of declared validity and not a measurement, whatever value it takes. Building that second branch, and asking whether a network coarse enough to fluctuate observably is compatible with the same three blocks, is the defined next step. It is named here as a task with an address, not as a prospect.

V | A pre-registered test of the inhomogeneous sector

This appendix fixes, before the computation is run, what will be computed, how the outcome will be read, and what will not be changed afterwards. It is dated and enters the deposited archive with the rest of the package, so that the record precedes the result.

Claim under test. The count noise of the local transition, negligible above about 10^{11} records per coherence cell at the reference coherence length of Appendix U, is not negligible below it. The test asks whether the three data blocks of Appendix S are compatible with a network coarse enough for that noise to matter, over the whole range the model admits, $10^4 \le N_{\mathrm{seed}} \le 10^{25}$.

Computation. The certificate already stores, for every design point, the linear response of every release row to each unit column of the count noise, at the three variances 10^{-4}, 2×10^{-4} and 10^{-3}, together with the closure coefficients that give the mean shift of each row per unit variance. The second branch assembles from them, for each sample, the mean shift $\mu(\theta)$ and the noise covariance, and replaces the omission bound of Appendix U by the full form

$$-2\ln\mathcal{L} = (r-\mu)^T(C + A\,\Sigma\,A^T)^{-1}(r-\mu) + \ln\det(C + A\,\Sigma\,A^T), \tag{V1}$$

where C is the released covariance of the three blocks, A the stored linear response and Σ the count-noise covariance, whose scale follows (U1). The rank of $A\Sigma A^T$ is at most the number of noise columns, so the inverse and the determinant are evaluated with the Woodbury identity already used for the supernova block. The emulator supplies A and μ where it supplies the certificate blocks now; its declared error enters (V1) as it enters the present likelihood.

Decision rule, fixed now. The chain is run with the prior of the present comparison, unchanged, and the strong reading $\nu_v = 0$. The marginal posterior of $\log N_{\mathrm{seed}}$ is read against a reference, since with a prior edge an interval can exclude the floor with no information from the data. The reference is a second chain run with the same sampler, the same prior and the same refusals, but a constant likelihood: a truncation or a refusal near the floor removes samples from both chains and cannot be read as a measurement. Let F be the lowest tenth of the admissible range, $4 \le \log_{10} N_{\mathrm{seed}} \le 6.1$. In each chain the fraction of draws in F becomes an effective count, with the larger of the autocorrelation times of $\log N_{\mathrm{seed}}$ and of the indicator of F, and one-sided 97.5 per cent Clopper-Pearson bounds are taken: U above the posterior fraction, L below the reference fraction; with no visits U stays above zero. On effective counts these bounds are approximate and do not guarantee exact coverage for correlated samples. The rule is deposited as a script, and the script, not this paragraph, will be applied.

Not assessable. With fewer than 500 effective draws in either chain, or fewer than 50 effective reference visits to F, the rule cannot be applied; the chains may be extended

once, to four times their length, and if that fails the test is inconclusive on that side, not an outcome either way.

First outcome. If $U/L < 0.1$, more than ten to one against a coarse network, conventionally strong evidence, the one-sided 95 per cent lower limit of N_{seed} is reported as the result.

Second outcome. Otherwise the pre-registered threshold is not reached. That is not a statement that the data carry no information on the lower tail: the one-sided 95 per cent lower limits of posterior and reference are both reported and compared, but no lower bound is claimed as the result of the test. The upper side is read in the same way on the highest tenth and reported separately; a coarse network would then be sought in the matter power spectrum of the coupled Boltzmann runs.

Checks of the rule, before deposit and again before the run. Six synthetic cases with autocorrelated chains are fixed now and deposited with their outcomes: a constant likelihood gives the second outcome, a likelihood vanishing below $\log_{10} N_{\mathrm{seed}} = 8$ the first, no support in F and chains too short are not assessable, a posterior mass of 0.001 in F gives the first, and a mass of 0.02, which moves the lower limit from 5.1 to 6.7 short of the threshold, the second. All six pass. If any fails when repeated before the run, the rule is not applied.

Third outcome. If the linearity test of the closure exceeds its 5 percent tolerance on any tested row, or the emulator error on A exceeds one tenth of the corresponding row standard deviation, the branch cannot be certified with this machinery, which is reported, not repaired by loosening either tolerance.

What will not be done. No prior is changed after the result is seen, no data block is added or removed, and the gate, the vacuum fraction and the thresholds above stay as written here. The weak reading $\nu_v > 0$ is a secondary analysis, labelled as such, and whatever the outcome it is added to this package in a dated revision.

Two statements the model already makes. Independently of the test, the branching law gives $\bar{w}_D \geq -1$ at every epoch, by (S1) and by the weighting with any constant fraction: the model cannot produce a phantom crossing. A confirmed crossing in a future release of the baryon-oscillation or supernova data, at a significance the release itself declares decisive, would exclude the model in this form. And under the strong reading Appendix T bounds p, the departure of the relational component from a cosmological constant, to a few hundredths: a larger measured departure, with the equation of state above -1 that (S1) requires, would exclude it too.

On the method. The protocol, the code and this text were prepared by the author with an AI assistant, which wrote and ran the scripts under his direction. That fact is stated for one reason: so that the reader treats the assistance as a method, subject to the same reproducibility as any other, and not as a source of authority in either direction. Every number in this package was produced by a script, and the test above will be judged by the scripts, not by the author or by the assistant.

A short glossary to return to the narrative

Relational link. A relational connection. In this article's mathematical instance, a channel between two registers with two possible states. It translates the Italian term *nesso*.

Register. An element capable of carrying a distinguishable state. The term presupposes neither conscious memory nor a human observer.

Fractionation. Attenuation of the comparison transmitted through a link. The share q is a correlation coefficient in the adopted channel, not a photon's survival probability.

Capacity. The total number of recording slots, including those inactive for the count being tracked. Its translation into volume is an additional physical assumption.

Redundancy. A statistical outcome that does not increase the active total in channel B. Its probability p is not the fraction q.

Distal. In this model, relating to accumulated interpositions between epochs being compared. It does not mean that a region has more energy today because it is far from an observer.

Comoving. Following the mean cosmic expansion. A comoving region retains its identity while its physical volume changes.

Scale factor. The number a describing the growth of physical distances between comoving references. Their distances in comoving coordinates remain constant. In three-dimensional geometry, physical volume varies as the cube of a.

Pressure-to-density ratio. The ratio $w = P/\rho$, with ρ denoting energy density. It expresses the contribution of pressure relative to density in the cosmological bridge.

Redshift. The relative stretching of wavelength, denoted by z. The stretching factor is $1 + z$.

Dominant regime. The part of the evolution in which the most persistent growth mode prevails. It need not coincide with the entire observed interval.

Covariance. A joint description of uncertainties: it also records how errors in different measurements are related.

Frozen parameter. A parameter retained at its earlier value in the new comparison. This does not mean it was derived from the theory or that the test is blinded.

Effective bridge. A set of identifications and physical laws adopted to translate the model into observable quantities, without claiming a fundamental derivation here.

Memory-space extension. The separate candidate model of I-R: spatial registers and memories have distinct roles. Its reading channel r is not the original epoch channel q.

References and related reading

Cosmology, data and neighbouring programmes

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Model provenance and supporting materials

[11] Ugo Lanciano, MM 2.30 unified volume, English edition 0.19, source record identified by the author: DOI 10.5281/zenodo.22730712. This article does not modify that record. The 2.32 package is a separate, restricted contribution, not a replacement for the unified volume.

[12] Working notes R30, *Candidate bridge between MM and the distal hypothesis*, and B14, *Links, counting and regularity* (September 2026; titles translated from Italian). They supply the capacity-volume bridge and the probabilistic realization of a link, respectively. The channel with counts $2n + 1$ and one, and the joint process in Appendix B, were developed in the first edition of this article.

[13] B17-R1 research manuscript, *Robust distal growth from a relational branching model* (13 September 2026). Source of the branching family, robustness region and structural checks used in the appendices. It is not independent observational confirmation. The 2.32 package retains it as a companion manuscript.

[14] Research report, *A restricted observational transfer test of the MM distal background* (13 September 2026). Source of the frozen parameters, DES selection, Table 1 and diagnostics in G-H. Its results are inherited here, not recalculated: the independent comparison of Appendices S-U uses a different supernova sample, Pantheon+ above redshift 0.01 [19], with its own background, its own likelihood and two further data blocks. The companion PDF reproduces the dated report with a packaging note.

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[18] Emulator and certificate materials for the comparison of Appendices S-U: exact background solution, HyRec recombination on the actual expansion history, the Gaussian-process emulator of the recombination and certificate blocks, its leave-one-out certification and the sampled chain. Internal materials of this package, not a publication.

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What accompanies the text

The separate research-materials archive contains the editable source, figures, internal mathematical checks, their results and the cited observational result files. **These are separate supporting files: this PDF has no embedded attachments.** Full supernova catalogues and the original training inputs are not included; the package does not claim standalone end-to-end observational replication. The calculation code that the profile likelihoods of Appendix T need has been collected, with scripts that download the three data blocks and HyRec-2 from their original sources, in a reproduction kit prepared and tested with this release. The kit is not part of this deposit; it will be deposited as a separate record that cites this one.

The documented standalone commands rerun mathematical checks of the formulated models, not their ontological origin. Mathematical proofs and numerical checks serve different purposes; the latter neither replace the former nor constitute independent review. The dated companion documents retain their original scope. Release preparation adds no fit to them; the new fit of Appendices S-U is reported separately, with its own machinery and its own limits.

The original 29-page English article translated the refined Italian source. This revision preserves its equations, cosmological parameters, samples and observational results, and adds the new narrative and Appendices I-R directly in English. The local Italian edition does not yet contain these additions. The English text is the reference edition for this package; translation-parity records apply only to the earlier common content.

The archive includes a portable English implementation of the homogeneous reading process in O-Q, its results and regeneration instructions, in addition to the ideal-chain checks. Older mesh evidence and the pair audit have explicit source fingerprints and scope limits; the original Italian prototypes are not a bundled portable implementation. Appendices S-U do add an observational comparison, performed by the author with AI assistance; no independent scientific validation is claimed for it.

The ontological architecture is proposed by the author; formalization and writing were developed with AI assistance under his direction. Translation and editorial preparation also used AI assistance. No claim of a second research group or independent scientific review is made. Author identifier: ORCID 0009-0008-7618-9635.